Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds*

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Compact Lie groups have discrete unitary representations (Peter–Weyl theorem); the GeoVac graph exploits this discreteness directly. Transcendental conversion factors appear when these discrete representations are projected onto non-compact coordinate systems used for measurement. We classify these factors by the type of compactness mismatch they mediate—a six-tier taxonomy (intrinsic, conformal/calibration, embedding, algebraic-implicit, composition, inner-factor input data) spanning the GeoVac hierarchy through Level 5 plus the almost-commutative inner factor—and identify them with Weyl–Selberg exchange constants of spectral geometry. The falsifiable claim: the transcendental content of any GeoVac observable is determined entirely by the type of projection linking the compact discrete description to the non-compact coordinate system.

Beyond the basis-level catalog, three theorems classify the spectral content of operators on round S^3: **(Theorem 1)** operator order is the primary discriminant of transcendental class—the squared Dirac operator D^2 produces π^{even} exclusively at every integer s, while the first-order Dirac operator D produces odd-zeta content $\zeta_R(2k{+}1) \perp \mathbb{Q}(\pi)$ at odd s; **(Theorem 2)** the full classification is three-axis, adding bundle type (scalar vs. spinor, modulating coefficients only) and diagram topology (free vs. vertex-restricted, introducing Dirichlet L-values $G = \beta(2)$, $\beta(4)$ through the quarter-integer Hurwitz shift); **(Theorem 3)** the three components of the fine-structure-constant combination $K = \pi(B + F - \Delta)$ of Paper 2 [7] have structurally independent spectral homes (finite Casimir trace, infinite Dirichlet series at the packing exponent, Dirac mode degeneracy on the spinor bundle), each in a distinct cell of the three-axis grid. The combination rule $K = \alpha^{-1}$ remains a numerical observation without first-principles derivation.

A complementary **Observation (structural incommensurability)** catalogues the meta-pattern: GeoVac observables that look unitary at the surface (K itself; two-loop QED on S^3; zero structure of the GeoVac zeta) resolve under structural analysis into objects from different cells of the three-axis grid. No further unification is currently visible. The taxonomy is identified as a classification of Mellin transforms of heat kernels: the free heat kernel produces π^{even}, the interaction-deformed kernel produces L-content via vertex parity, and operator order determines whether the Mellin integrand samples squared or raw eigenvalues. The classification is R-independent, persisting in the flat-space limit via Weyl's law.

I. INTRODUCTION

The Geometric Vacuum program constructs discrete lattice Hamiltonians by identifying the natural geometry of each quantum system and discretizing it as a sparse graph [1–3]. A persistent observation across the hierarchy of natural geometries is that the discrete algebraic structures—integer eigenvalues, finite recurrence relations, Gaunt integrals built from Wigner $3j$ symbols—are entirely self-contained, but that their connection to physical observables in continuous coordinate systems always introduces transcendental constants.

The thesis of this paper is that these transcendental conversion factors are consequences of *compactness.* Compact Lie groups have discrete unitary representations (the Peter–Weyl theorem [32]); the GeoVac graph exploits this discreteness directly. Transcendental numbers enter when—and only when—those discrete representations are translated into non-compact coordinate systems used for measurement. The taxonomy developed below classifies the resulting conversion factors by the type of compact-to-non-compact projection that produces them. This yields a falsifiable structural claim: the transcendental content of any GeoVac observable is determined entirely by the type of projection linking the compact discrete description to the non-compact coordinate system (Claim 4, Sec. VI).

The simplest instance is already visible in Paper 0's packing construction [1]. Axiom 2 (geometric indifference) forces states onto isotropic shells—circles, i.e. copies of S^1. The shell is compact; its circumference $2\pi r$ introduces π into the area formula $\sigma_0 = \pi d_0^2/2$ and hence into the state count $N_k = A_k/\sigma_0 = 2(2k-1)$. The integer state count is a consequence of compact topology; π is the price of expressing that count as a ratio of continuous areas. This pattern—compact geometry producing integer quantum numbers, with π entering through projection onto non-compact coordinates—recurs at every level of the hierarchy.

The conversion factors are instances of Weyl's law [27] and the Selberg trace formula [28]; what is new is their systematic identification across the GeoVac hierarchy, culminating in a falsifiable observable classification (Claim 4) and the observation that operator order on the compact space determines the motivic weight of the

* Paper 18 in the Geometric Vacuum series

resulting transcendental content.

II. CATALOG OF EXCHANGE CONSTANTS

A. Level 1: $\kappa = -1/16$ ($S^3 \to \mathbb{R}^3$)

The compactness of S^3 is what makes the graph eigenvalues discrete integers; κ converts this discrete spectrum to the non-compact $\mathbb{R}^3$ energy scale.

The continuum S^3 Laplace–Beltrami operator — which the (positive-semidefinite) graph Laplacian converges to — has pure integer eigenvalues

$$\lambda_n = -(n^2 - 1), \qquad n = 1, 2, 3, \ldots \tag{1}$$

with degeneracy $g_n = n^2$ [3]. The physical hydrogen energy levels $E_n = -1/(2n^2)$ are related by the energy-shell constraint $p_0^2 = -2E$, which acts as a stereographic focal length projecting S^3 onto flat $\mathbb{R}^3$. The universal kinetic scale factor

$$\kappa = -\frac{1}{16} \tag{2}$$

calibrates the ground state energy: $\kappa \cdot \lambda_{\max} = E_H = -1/2$ Ha, where $\lambda_{\max} \to 8$ is the maximum graph Laplacian eigenvalue in the continuum limit [1]. Per-shell energies $E_n = -Z^2/(2n^2)$ emerge from the full graph Hamiltonian $H = \kappa\mathcal{L} + W$ (with node weights $W_{nn} = -Z/n^2$), not from κ alone.

The geometric quantity that κ's value *coincides* with is well understood: it is the Jacobian of the stereographic projection composed with the conformal factor of the Fock map. (That the calibration value κ coincides with this geometric factor is an Observation, not a derivation — no bridge derives the calibration role from the geometry; see §VII F.) Specifically, if Ω is the conformal factor $\Omega = 2p_0/(p^2 + p_0^2)$ (so that $\Omega(0) = 2$ at the ground-state shell $p_0 = 1$, matching §VII F and Paper 7 [3]), then the Laplace–Beltrami operator on S^3 and the flat-space kinetic operator are related by $\nabla^2_{S^3} = \Omega^{-2}[\nabla^2_{\text{flat}} + \text{lower order}]$, and κ absorbs the accumulated conformal rescaling and the p_0 energy-shell normalization [3].

The key property of κ is that it is *unnecessary on the graph*. The graph Laplacian eigenvalue problem $L\psi = \lambda\psi$ is solved entirely in terms of integers. The constant κ appears only when the result is translated into atomic units (Hartrees, bohrs)—i.e., when the discrete spectrum is projected onto a continuous coordinate system.

B. Level 2: $e^a E_1(a)$ (Laguerre basis → prolate spheroid)

The Laguerre spectral basis inherits the compact angular structure of S^3; the Stieltjes seed $e^a E_1(a)$ enters when this basis is embedded in the non-compact prolate spheroidal coordinate, whose curvature introduces a pole absent from the compact description.

The prolate spheroidal radial solver uses a Laguerre spectral basis $\varphi_n(\xi) = e^{-\alpha(\xi-1)} L_n(2\alpha(\xi - 1))$ with matrix elements computed from three-term recurrences [4]. For σ states ($m = 0$), all matrix elements are purely algebraic: the Laguerre moment matrices $M_k[i,j] = \int_0^\infty x^k L_i(x) L_j(x) e^{-x} dx$ are band-structured matrices with rational entries, built from the recurrence $xL_n = -(n+1)L_{n+1} + (2n+1)L_n - nL_{n-1}$.

For $m \neq 0$ states, the centrifugal term $m^2/(\xi^2 - 1)$ introduces a $1/x$ singularity in the Laguerre variable $x = 2\alpha(\xi - 1)$. The associated Laguerre basis $L_n^{|m|}(x)$ with weight $x^{|m|}e^{-x}$ absorbs this singularity by construction. The centrifugal matrix elements decompose via partial-fraction expansion:

$$\frac{m^2}{\xi^2 - 1} = 4\alpha^2 m^2 \left[\frac{1}{x} - \frac{1}{x + 4\beta}\right], \tag{3}$$

where $\beta = \alpha(\xi_0 - 1)$ is the Laguerre scale offset. The two terms have fundamentally different algebraic character:

- The $1/x$ term produces a lowered moment $M_{-1}[i,j] = \int_0^\infty x^{|m|-1} L_i^{|m|}(x) L_j^{|m|}(x)\, e^{-x}\, dx$, which is fully algebraic via the DLMF 18.9.13 summation identity [26].
- The $1/(x + 4\beta)$ term produces a Stieltjes matrix $J[i,j] = \int_0^\infty \frac{x^{|m|}}{x+a} L_i^{|m|}(x) L_j^{|m|}(x)\, e^{-x}\, dx$, where $a = 4\beta$. This matrix satisfies a three-term recurrence in the row/column indices, seeded by the single transcendental

$$S_0^{(\alpha)}(a) = \Gamma(\alpha + 1)\, e^a\, E_{\alpha+1}(a), \tag{4}$$

where $E_n(a) = \int_1^\infty t^{-n} e^{-at}\, dt$ is the generalized exponential integral and $a = 4\beta$ is the Laguerre scale parameter [20]. For integer $\alpha = |m|$, all higher E_n reduce to $E_1(a)$ via a finite rational recurrence, so the entire matrix is built from algebraic operations plus one transcendental constant.

The computational cost of the transcendental seed is negligible: one call to $E_1(a)$, which achieves machine precision for $a \geq 4$ (covering all physical parameters). The seed then propagates through an $O(N^2)$ algebraic recurrence that produces all matrix elements. The continuous mathematics has been compressed from a *function* (the quadrature integrand over $[0, \infty)$) to a *constant* (the Stieltjes seed), with the entire function-level information reconstructed algebraically from that constant.

An unexpected computational bonus: the associated Laguerre basis converges *faster* than the ordinary Laguerre basis for $m \neq 0$ states. For $|m| = 1$ (π states), the associated basis is stable by $N = 10$, compared to non-monotonic convergence at $N \approx 20$ for the ordinary basis. For $|m| = 2$ (δ states), algebraic-quadrature agreement reaches $\sim 5 \times 10^{-9}$ (machine precision), with

PES shape (equilibrium geometry) preserved exactly [20]. This suggests that identifying the exchange constant and absorbing it into the basis does not merely explain the transcendence—it *improves the spectral method*, because the adapted basis eliminates the curvature mismatch that the exchange constant mediates.

The Stieltjes seed $e^a E_1(a)$ is the Laplace transform of the rational function $1/(1+t)$ evaluated at a. It encodes how the Laguerre inner product (defined on the flat half-line $[0,\infty)$ with exponential weight) "sees" a shifted pole at $x=-a$ arising from the curvature of the prolate spheroidal coordinate system. On the graph, there is no pole; the centrifugal barrier is a selection rule encoded in the quantum numbers. The transcendental appears only when this selection rule is represented as a rational function in a continuous coordinate.

1. The full Level-2 seed set: $\{e^a E_1(a), \ln, \gamma_E\}$

Track J isolated $e^a E_1(a)$ for the *one*-electron problem at $m \neq 0$. The two-electron two-centre problem — the electron repulsion integrals over prolate spheroidal coordinates — is the same projection carried further, and it is worth recording what the complete seed set turns out to be, because the answer is smaller than one might fear and its two negatives are load-bearing.

A native closed-form evaluation of all four two-centre ERI classes (Paper 58) gives seeds

Class	Seeds	Weight
$(AA\|BB)$	— (elementary)	0
$(AA\|AB)$, $(AB\|BB)$	$e^a E_1(a)$, ln	1
$(AB\|AB)$	$e^a E_1(a)$, ln, γ_E	1

The seed set *grows* with class difficulty; the transcendence *weight* does not. Each new object has a structural home rather than being an independent import. The logarithm is the Neumann kernel $Q_0(\xi) = \frac{1}{2}\ln[(\xi+1)/(\xi-1)]$, i.e. the same prolate spheroidal curvature seen from the kernel side instead of the moment side; γ_E is not independent of E_1 at all, arriving with it through $E_1(z) = -\gamma_E - \ln z + \mathrm{Ein}(z)$. All three therefore sit in the *embedding* tier alongside $1/r_{12}$: they are the cost of the continuum two-centre geometry, not content of the graph.

Two absences matter more than the list. First, **no π survives**: it enters through the spherical normalisations and cancels identically against the 4π of the Coulomb kernel, so no calibration-tier constant appears anywhere in a two-centre ERI, and the master Mellin engine's $M_1/M_2/M_3$ sub-mechanisms are simply not in this chain. Second, **no weight-two object appears** — no dilogarithm, no $\zeta(2)$ — although the ordered $\xi_</\xi_>$ integral is formally an iterated integral over a simplex, exactly the shape that generically produces Li_2. It closes at weight one instead.

The two absences together are what make the $(AA|BB)$ class *decidable*. With no π and no weight-two content the closed form is $A_0(R) + \sum_j A_j(R)e^{-\lambda_j R}$ with A_j, λ_j rational, so by Lindemann–Weierstrass the entry vanishes if and only if every A_j does — a decision, not a numerical threshold. This is the sense in which cataloguing a transcendental *buys* something rather than merely labelling it: knowing the exact seed set is what licenses an exact zero test downstream.

2. Three centres: the embedding tier is genus-graded

The two-centre weight-one closure is a *genus-zero* fact, and it stops at two centres. The two-body three-centre integral $(XY|XZ)$—the first object a polyatomic requires, and one no prolate spheroidal system can hold, that coordinate having exactly two foci—is naturally evaluated in momentum space, where the Coulomb kernel is $4\pi/k^2$ and each centre enters as a phase $e^{i\mathbf{k}\cdot\mathbf{R}}$ rather than a focus. There the two densities Fock-project onto spheres of *different* radii $\sqrt{c_1} \neq \sqrt{c_2}$ (c_i the Feynman parameters of the two densities), and the product of the two stereographic weights is the quartic

$$y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1), \tag{5}$$

an elliptic curve. Its $D=0$ period is a complete elliptic integral K, and the full integral is an elliptic polylogarithm on this family, degenerating to the genus-zero two-centre curve only on the measure-zero locus $c_1 = c_2$ of a shared sphere.

The embedding tier therefore carries a *genus grading*: genus zero ($\{e^a E_1(a), \ln, \gamma_E\}$) at two centres, genus one (elliptic) at three. The two two-centre negatives above—no π, no dilogarithm—are genus-zero properties and do *not* survive the third centre, which does not climb to a higher rung of the polylogarithm/MZV weight tower but leaves that tower entirely, orthogonally to the calibration $M_1/M_2/M_3$ Mellin sub-mechanisms. This is the transcendence-level statement of "no coordinate system for three foci," and the reason Gaussian orbitals—whose product theorem collapses three centres to one—displaced Slater orbitals for polyatomic quantum chemistry. The elliptic identification is numerical (the $D=0$ period verified to 31 digits against a momentum-space evaluation that reproduces the exact three-centre integral to $\sim 10^{-14}$); the closed form in elliptic polylogarithms is not derived here.

C. Level 3: $\mu(R)$ ($S^3 \times S^3 \to$ hyperspherical)

On the compact product space $S^3 \times S^3$, the two-electron Hamiltonian has purely algebraic matrix elements; the adiabatic eigenvalue $\mu(R)$ enters only when this compact description is reparameterized into the noncompact hyperspherical coordinate R.

The two-electron problem on $S^3 \times S^3$ is fully algebraic: the Hamiltonian matrix has entries given by Gaunt integrals (finite sums of Wigner $3j$ symbols with rational coefficients), and the eigenvalues are roots of a polynomial with algebraic coefficients. The Full CI solver exploits this, achieving 0.35% error for helium with no transcendentals anywhere [6].

The hyperspherical reparameterization introduces coordinates (R, α) where $R = \sqrt{r_1^2 + r_2^2}$ is the hyperradius and $\alpha = \arctan(r_2/r_1)$ is the hyperangle. The angular eigenvalues $\mu(R)$ depend parametrically on R because the effective angular Hamiltonian is $H(R) = H_0 + R \cdot V^{\text{coupling}}$ [5].

Each perturbation coefficient a_k in the expansion $\mu(R) = \mu^{(0)} + a_1 R + a_2 R^2 + \cdots$ has a closed form: the a_k are finite sums of algebraic numbers (square roots of rationals from Clebsch–Gordan coefficients) times powers of π^{-1} (from normalization over the angular domain $[0, \pi/2]$), arising from partial harmonic sums over SO(6) Casimir eigenstates [5]. For example, $a_1(n{=}1, Z) = (8/3\pi)(\sqrt{2} - 4Z)$. However, the full function $\mu(R)$ is not transcendental in the sense originally claimed. The angular Hamiltonian $H(R) = H_0 + R \cdot V_C$ is a linear matrix pencil, so its eigenvalues satisfy the characteristic polynomial

$$P(R, \mu) \;=\; \det\bigl(H_0 + R\, V_C - \mu\, I\bigr) = 0, \qquad (6)$$

which is a polynomial of degree d in both R and μ (where d is the matrix dimension at a given truncation). The coefficients of P lie in $\mathbb{Q}(\pi, \sqrt{2})$, since H_0 has integer entries (SO(6) Casimir eigenvalues) and V_C has entries involving π^{-1} and $\sqrt{2}/\pi$ from the angular integrals. This makes $\mu(R)$ an *algebraic* function of R over the coefficient ring $\mathbb{Q}(\pi, \sqrt{2})$—it is a root of a polynomial, not a transcendental function [23].

The transcendental content of $\mu(R)$ is therefore *inherited* from the coupling matrix V_C, which contains upstream exchange constants of embedding type (π^{-1} from angular normalization, $\sqrt{2}$ from Gaunt-type integrals). The R-dependence itself introduces no new transcendentals: it is purely algebraic.

The failure of the Taylor/Padé expansion at large R (Track G [21]) is reinterpreted as a *branch point* phenomenon, not evidence of transcendence. The algebraic curve $P(R, \mu) = 0$ has branch points where eigenvalue sheets meet—these are the zeros of the discriminant $\Delta(R) = \mathrm{disc}_\mu(P)$. The Taylor series around $R = 0$ has a convergence radius set by the nearest branch point in the complex R-plane. Padé approximants, being rational functions, also cannot cross branch points where $\mu(R)$ has square-root type singularities. But the implicit relation $P(R, \mu) = 0$ is exact for *all* R, including $R > 5$ where the series diverges. This has been verified computationally to machine precision at $R = 50$ bohr [23].

The $O(R) \to O(R^2)$ crossover, which was previously interpreted as a regime change requiring transcendental interpolation, is simply the behavior of an algebraic function whose dominant term changes character: at small R, the linear term $V_C \cdot R$ dominates; at large R, higher-order terms in the characteristic polynomial dominate. The crossover is smooth along the algebraic curve, even though it is singular from the perspective of a Taylor expansion centered at $R = 0$.

The crucial point remains that $\mu(R)$ is not intrinsic to the two-electron problem. It is an artifact of the hyperspherical coordinate system. The $S^3 \times S^3$ product space does not require $\mu(R)$; it only appears when one reparameterizes for computational efficiency. The exchange rate between the discrete product-space description and the continuous hyperspherical description is an algebraic function of R, with transcendental content inherited from the angular coupling integrals.

D. Level 4: $\mu(\rho, R)$ (product space → molecule-frame hyperspherical)

At Level 4 the compact product basis (prolate spheroidal CI) is fully algebraic but structurally incomplete; the non-compact mol-frame hyperspherical reparameterization introduces $\mu(\rho, R)$ as the irreducible cost of accessing three-body coalescence physics that the compact description cannot represent.

Level 4 combines the two-center complication of Level 2 with the two-electron complication of Level 3. Two approaches exist, with fundamentally different algebraic character.

The product-space approach (prolate spheroidal CI, Paper 12) is fully algebraic: all electron-electron matrix elements are computed from Neumann expansion recurrence relations with zero spatial quadrature. The Laguerre radial and Legendre angular matrix elements are likewise algebraic. No transcendental constant appears anywhere in the computation.

However, a systematic convergence study (Track M) demonstrates that this algebraic approach saturates at ∼92.5% of the exact dissociation energy D_e [22]. The angular basis convergence is rapid—92.3% at $l_{\max} = 2$ (46 basis functions), 92.5% at $l_{\max} = 4$ (114 basis functions)—but successive improvements decay by a factor of 200 from $l = 1 \to 2$ to $l = 2 \to 3$, with the $l = 3 \to 4$ increment contributing only 0.04%. Extrapolation in the slow regime predicts that 5% error would require $l_{\max} \sim 45$ and 1% error $l_{\max} \sim 113$, both impractical. The effective ceiling is 92.5% D_e.

The smoking gun is explicit correlation: 9 basis functions with r_{12}^2 factors (Hylleraas $p = 2$) achieve 94.7% D_e, exceeding the 114-function prolate CI plateau. The residual 7.5% matches Paper 12's cusp diagnosis (7.6%): the electron-electron cusp ($1/r_{12}$ singularity) is a completeness deficiency of the product basis, not a convergence issue. The transcorrelated approach (Paper 14, Sec. IV; Paper 15, Sec. VI.J) removes this singularity via a Jastrow similarity transformation, converting $1/r_{12}$ from a hard embedding constant to a smooth gradient operator at the cost of non-Hermiticity (Sec. IV).

The mol-frame hyperspherical reparameterization (Paper 15) resolves this by introducing coordinates (ρ, R, α)

where ρ is the electron hyperradius and α the hyperangle. The angular eigenvalues $\mu(\rho, R)$ are now a transcendental function of *two* continuous parameters, reflecting the simultaneous three-body coalescence in coordinate and configuration space. This solver achieves 94.1% D_e, surpassing the prolate CI ceiling.

The distinction between Levels 3 and 4 is qualitative. At Level 3, the $S^3 \times S^3$ FCI converges (albeit slowly)—the exchange constant $\mu(R)$ is a computational convenience that accelerates convergence but is not fundamentally necessary. At Level 4, the prolate product-space CI *saturates*: the exchange constant $\mu(\rho, R)$ is irreducible because the product basis is structurally blind to three-body coalescence. The reparameterization does not merely accelerate convergence; it accesses physics that the algebraic product space cannot represent at any practical basis size.

E. Level 5: Spectral-side exchange constants on S^3

Four further realizations of the taxonomy live on the same S^3 substrate as Levels 1–4 but classify projections of *spectral* rather than basis-level data: a closed-form Dirichlet L-content identity in the Dirac Dirichlet difference; a GUE/Poisson dichotomy between the spectral-zeta and Ihara-zeta sides of the framework; an algebraic-but-non-radical tier produced by a non-solvable Galois factor of the Ihara zeta on S^5; and an α^2-weighted Ihara zeta living in the ring $\mathbb{Q}(\alpha^2)[s]$. All four are catalogued here as new entries in the basis-level catalog of Levels 1–4; their meta-pattern (categorical disagreement between spectral and combinatorial views of the same Dirac data) is treated in Sec. VIII.

a. Closed-form χ_{-4} content in the Dirac Dirichlet difference. On S^3 with the Camporesi–Higuchi Dirac spectrum $|\lambda_n| = n+3/2$ and degeneracy $g_n = 2(n+1)(n+2)$, define the Dirac Dirichlet series $D(s) = \sum_n g_n/|\lambda_n|^s$ and its vertex-parity split $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$ (Paper 28 [12], eqs. (3)–(5)). The difference has the exact closed form

$$D_{\text{even}}(s) - D_{\text{odd}}(s) \;=\; 2^{s-1}\left(\beta(s) - \beta(s-2)\right), \quad s \in \mathbb{Z}_{\geq 2}, \tag{7}$$

where $\beta(s) = L(s, \chi_{-4})$ is the Dirichlet L-function of the mod-4 character. Two elementary Hurwitz identities drive the collapse: $\zeta(s, 5/4) = \zeta(s, 1/4) - 4^s$ and $\zeta(s, 3/4) = \zeta(s, 1/4) - 4^s\beta(s)$. Verified by sympy symbolic identity at $s \in \{2, \ldots, 10, 15\}$ and by PSLQ at 100-digit precision (`debug/spectral_chi_neg4_memo.md`). Eq. (7) is the concrete realization of Paper 28's qualitative "vertex parity acts as χ_{-4}" statement: the Dirichlet L-content enters the spectral-Dirichlet difference as a closed-form linear combination of $\beta(s)$ and $\beta(s-2)$ with integer prefactor 2^{s-1}.

b. RMT-spacing without critical-line confinement: the spectral–Ihara dichotomy. The zeros of $D(s)$, $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$ in the strip $\text{Im}(s) \in [0, 70]$ have unfolded imaginary-part spacing coefficient of variation CV $\in \{0.348, 0.343, 0.396\}$ —consistent with GUE (CV $\approx$ 0.46) and inconsistent with Poisson (CV $\approx$ 1.05). Kolmogorov–Smirnov testing rejects Poisson at $p = 0.004, 0.030$ respectively and cannot reject GUE. This is the first GUE signature in the GeoVac framework, and it is confined to the *spectral* Dirichlet side only: the combinatorial / Ihara-zeta side (graph non-backtracking walks) has Poisson / Berry-Tabor spacing at CV ≈ 1, a categorical dichotomy. Equally important, the $D(s)$ zeros are *not* on a single critical line (they are scattered in $\text{Re}(s) \in [2.02, 3.42]$), so the GUE signature is a spacing-only statement, not an RH-style line constraint: classical ζ has *both* RMT spacing and critical-line confinement; GeoVac's spectral zeta has only the first. This creates a new taxonomic tier, *RMT-spacing without critical-line confinement.*

c. Algebraic-but-non-radical transcendental content. The integer-coefficient Ihara-zeta factorizations produced in Paper 29 [13] factor over specific number fields as follows. Cyclotomic factors Φ_3, Φ_4, Φ_6 have Galois group $\mathbb{Z}/2$ (abelian); they fingerprint specific graph cycles (e.g. $\Phi_3, \Phi_6 \leftrightarrow$ 3-prism cycles of the $\ell = 1$ Coulomb S^3 component). Ramanujan-boundary factors $(4s^2+1)^2$ and $(9s^2+1)^4$ live in $\mathbb{Q}(i)$, consistent with the Hopf $\mathbb{Z}_2$ m-reflection sub-action of the Paper 25 Hopf $U(1)$ (see Observation 2 of [13]). *Bulk* factors have non-abelian Galois groups: S_3 for scalar S^3 cubics; S_4 for the Dirac-S^3 Rule-A $n_{\max} = 3$ quartic; and crucially S_6 (non-solvable) for the factor $P_{12}(s^2)$ in the scalar S^5 $N_{\max} = 3$ Ihara zeta. By Abel–Ruffini, the 12 zeros of P_{12} have *no radical expression* over $\mathbb{Q}$; by Kronecker–Weber, they do not lie in any cyclotomic extension $\mathbb{Q}(\zeta_M)$. This places a new tier in Sec. IV's output-class axis: *algebraic but not radical* — intermediate between closed-form radical algebraic and classical periods.

d. α^2-weighted Ihara zeta in the ring $\mathbb{Q}(\alpha^2)[s]$. The unweighted Ihara zeta depends only on 0/1 graph connectivity. Introducing the edge weighting $w(a, b) = 1 + \alpha^2 \cdot |f_{\text{SO}}(a) + f_{\text{SO}}(b)|/2$, where $f_{\text{SO}}(n, \kappa) = -(\kappa + 1)/[4n^3\ell(\ell+1/2)(\ell+1)]$ is the dimensionless Breit–Pauli spin-orbit diagonal, gives a weighted Ihara zeta $\zeta_G^w(s)$ living in the ring $\mathbb{Q}(\alpha^2)[s]$ — an explicit realization of the spinor-intrinsic tier without the $\gamma = \sqrt{1-(Z\alpha)^2}$ radical (`geovac/ihara_zeta_weighted.py`; the module's 14 tests cover the weighted-Ihara infrastructure, not the degree-44 / $-35/648$ values below, which are module-computed but not yet regression-tested — a coverage gap logged in `docs/claim_test_matrix.md`). For Rule-A at $n_{\max} = 3$, the inverse zeta expands as a polynomial in α of degree 44, with α^2 coefficient carrying the clean rational prefactor $-35/648$. The coefficient is a graph-invariant, not an edge-local perturbation — a concrete α^2-content projection.

Taxonomic placement of the Level-5 realizations

The four Level-5 constants slot into the existing taxonomy as follows. The χ_{-4} identity (7) is a *flow* exchange constant (infinite spectral sum), with the novelty that its value is an integer-coefficient linear combination of classical Dirichlet L-values, so Paper 18's transcendental taxonomy gains a "Dirichlet-L-closed" cell at the intersection of flow and Dirac-spectrum projection. The GUE-vs-Poisson dichotomy is a *meta-level* discriminant, not itself a transcendental, but classifies spectral objects by zero statistics — a new axis in Sec. IV. The algebraic-but-non-radical tier from P_{12} is a new *intrinsic* exchange-constant sub-class between Level 0 (rational/integer) and classical periods; it is the first time a non-solvable Galois group has been exhibited as a framework-native transcendental class. The α^2-weighted zeta is a *spinor-intrinsic* exchange constant in the ring $\mathbb{Q}(\alpha^2)[s]$, a subring of the full spinor-intrinsic tier $\mathbb{Q}(\alpha^2)[\gamma]/(\gamma^2 + (Z\alpha)^2 - 1)$ introduced in Sec. IV.

III. THE PATTERN: WEYL–SELBERG EXCHANGE

A. Weyl's law

The mathematical content of the observations in Sec. II is an instance of Weyl's law, which states that the eigenvalue counting function $N(\lambda)$ of the Laplace–Beltrami operator on a compact Riemannian manifold of dimension d satisfies [27]

$$N(\lambda) \sim \frac{\omega_d}{(2\pi)^d}\,\mathrm{vol}(\Omega)\,\lambda^{d/2} \qquad (\lambda \to \infty), \tag{8}$$

where $\omega_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit ball in $\mathbb{R}^d$.

Weyl's law relates two fundamentally different descriptions of the same manifold:

- The **spectral side**: $N(\lambda)$, a discrete staircase function counting integer-labeled eigenvalues.
- The **geometric side**: $\mathrm{vol}(\Omega)$, a continuous quantity measured by integration over the manifold.

The conversion factor $\omega_d/(2\pi)^d$ involves $\pi^{d/2}$—a transcendental that depends only on the dimension. This factor is the prototypical *spectral-geometric exchange constant*: it converts between counting (discrete) and measuring (continuous).

For the specific manifolds in the GeoVac hierarchy, the ball volumes ω_d carry $\pi^{\lfloor d/2 \rfloor}$:

- S^1 ($d=1$): $\omega_1 = 2$, Weyl constant $\omega_1/(2\pi)^1 = 1/\pi$.
- S^2 ($d=2$): $\omega_2 = \pi$, Weyl constant $\omega_2/(2\pi)^2 = 1/(4\pi)$.
- S^3 ($d=3$): $\omega_3 = 4\pi/3$, Weyl constant $\omega_3/(2\pi)^3 = 1/(6\pi^2)$.
- S^5 ($d=5$): $\omega_5 = 8\pi^2/15$, Weyl constant $\omega_5/(2\pi)^5 = 1/(60\pi^3)$.

In each case, the Weyl constant involves inverse powers of π (specifically $\pi^{-\lceil d/2 \rceil}$), reflecting the transcendental cost of converting a discrete eigenvalue count into a continuous volume measure.

B. Compactness as the source of discreteness

The Peter–Weyl theorem [32] states that the unitary representations of a compact Lie group form a discrete, countable, orthogonal decomposition of $L^2(G)$. This is why quantum numbers are integers: SO(4) is compact, so its irreps are labeled by discrete indices (n, l, m) with finite multiplicities $g_n = n^2$. Exchange constants arise when these discrete representations are expressed in non-compact coordinate systems—the stereographic $S^3 \to \mathbb{R}^3$ introduces κ; embedding in prolate spheroidal coordinates introduces $e^a E_1(a)$; reparameterizing $S^3 \times S^3$ as hyperspherical introduces $\mu(R)$. The transcendental content is the toll for crossing the compactness boundary.

C. π as the founding example

The simplest spectral-geometric exchange constant is π itself. The Laplacian on S^1 (the circle) has eigenvalues n^2, counted by integers; the continuous circumference is 2π. The conversion between the discrete eigenvalue count and the continuous measure is mediated by π. More generally, the volume of the unit d-sphere involves $\pi^{d/2}$, and the Weyl constant $\omega_d/(2\pi)^d$ carries exactly this factor. Every exchange constant in the GeoVac hierarchy is a descendant of this primordial example: κ inherits from the S^3 Weyl constant, the Stieltjes seed involves the exponential integral (a function whose Taylor coefficients involve the Euler–Mascheroni constant, itself a limit of harmonic sums weighted by π-dependent terms), and $\mu(R)$ is an eigenvalue of a Hamiltonian on S^5 whose Weyl asymptotics carry $\pi^{5/2}$.

Observation 1 (The graph is π-free)**.** *Every algebraic operation in the GeoVac graph construction produces integers or rationals:*

- *The Laplace–Beltrami spectrum $\lambda_n = -(n^2-1)$ that the graph Laplacian converges to: integer for all n.*
- *Degeneracies: $g_n = n^2$, integer.*
- *Gaunt integrals (Legendre normalization, $\int P_{l_1} P_k P_{l_2}\,dx$): ratios of factorials via Wigner 3j symbols, rational.*
- *Selection rules: combinatorial (triangle inequality on angular momentum labels).*
- *Clebsch–Gordan coefficients: algebraic numbers (square roots of rationals).*

- *Casimir eigenvalues:* $l(l+1)$, $\nu(\nu+4)/2$, *integer or half-integer for all levels.*

Transcendental numbers (π, $e^a E_1(a)$, $\zeta(2)$) *enter exclusively through normalization to continuous measures:* $\int |Y|^2\, d\Omega = 1$ *(with* $d\Omega$ *carrying* 4π*),* $\mathrm{vol}(S^3) = 2\pi^2$, *and spectral zeta functions. The exchange constants are precisely the* π*-injection points.*

D. The Selberg trace formula

Weyl's law is asymptotic. The Selberg trace formula [28] provides the exact identity:

$$\sum_j h(r_j) = \frac{\mathrm{Area}(\Gamma\backslash\mathcal{H})}{4\pi} \int_{-\infty}^{\infty} r\, \tanh(\pi r)\, h(r)\, dr + \sum_{\{\gamma\}} \cdots \tag{9}$$

where the left side sums over the discrete spectrum, the first term on the right involves the continuous volume (with a factor of π), and subsequent terms sum over conjugacy classes (geodesics). The Selberg trace formula is the bridge between the spectral world and the geometric world, and the transcendental factors (π, tanh, logarithms of algebraic numbers) are the toll for crossing.

When the quotient $\Gamma\backslash G$ is compact, the spectrum is purely discrete and the trace formula involves only algebraic data plus π-dependent conversion factors. When the quotient is non-compact, a continuous spectrum appears and the exchange involves Eisenstein series [28]. This parallels the GeoVac hierarchy: Level 1 (compact S^3) requires only a constant (κ); Level 3 involves an algebraic function $\mu(R)$ with transcendental coefficients (Sec. VII E).

E. The Mellin transform as the taxonomic engine

The preceding subsections identify the *what*—compact-to-non-compact projection introduces transcendental content—and the *where*—Weyl's law and the Selberg trace formula are the structural bridges. Here we identify the *how*: the Mellin transform of the heat kernel trace is the single mathematical operation that converts the discrete spectral data of a compact manifold into the transcendental content classified in Sec. IV.

For a non-negative self-adjoint operator A on a compact Riemannian manifold, the spectral zeta function is related to the heat kernel trace by [40, 41]

$$\zeta_A(s) \;=\; \frac{1}{\Gamma(s)} \int_0^{\infty} t^{s-1}\; \mathrm{Tr}\big(e^{-tA}\big)\, dt\,, \tag{10}$$

where $\mathrm{Tr}(e^{-tA}) = \sum_n g_n\, e^{-t\lambda_n}$ is a purely spectral object: the sum runs over the discrete eigenvalues λ_n with their integer degeneracies g_n. Both sides of Eq. (10) encode the same information—the spectrum of A—but on the left in a form that produces transcendental numbers ($\pi^{d/2}$ from the Seeley–DeWitt asymptotic expansion of the heat trace, odd-zeta values from the analytic continuation), and on the right in a form that is manifestly a function of integer-labeled eigenvalues and degeneracies. The Mellin transform is the bridge; the $\Gamma(s)$ prefactor is the conversion toll.

a. Operator order determines the Mellin output class. The three-axis classification of QED transcendentals on S^3 (Sec. IV) has its foundation in the distinction between applying the Mellin transform to the squared eigenvalues λ_n^2 versus the raw eigenvalues $|\lambda_n|$.

For the *squared* Dirac operator D^2 on unit S^3, the eigenvalues are $(n+3/2)^2$, which are perfect squares of half-integers. The heat trace $\mathrm{Tr}(e^{-tD^2})$ is a theta-function-like object whose Mellin transform samples only even powers: the variable substitution $m = 2n+3$ converts the spectral sum to a sum over odd integers, and the Euler–Bernoulli formula for $\zeta_R(2k)$ maps every term to rational $\times\, \pi^{2k}$. This is the content of the T9 theorem (Paper 28, Theorem 1 [12]): $\zeta_{D^2}(s)$ is a two-term polynomial in π^2 at every integer s. No odd-zeta value can appear because the squaring operation has removed the odd-power structure from the Mellin integrand.

For the *first-order* Dirac operator $|D|$ with eigenvalues $n+3/2$, the Mellin transform accesses the raw half-integer spectrum. The Dirac Dirichlet series $D(s) = \sum g_n |\lambda_n|^{-s}$ involves both even and odd powers of the spectral parameter, and the parity discriminant (Paper 28, Theorem 2 [12]) follows directly: at even s, both constituent Hurwitz zeta terms are Bernoulli-type (π^{even}); at odd s, both involve genuinely independent odd-zeta values. The operator order—first versus second—determines whether the Mellin transform samples the raw spectrum or its square, and this single distinction generates the first axis of the three-axis classification.

b. Interaction deforms the heat kernel. The Coulomb potential $V = -Z/r$ perturbs the free Laplacian on S^3. In the heat kernel framework, the perturbed trace $\mathrm{Tr}(e^{-t(A+V)})$ has an asymptotic expansion whose Seeley–DeWitt coefficients $a_k(A+V)$ differ from the free coefficients $a_k(A)$ by terms involving the potential and its derivatives [42]. The Mellin transform of this perturbed heat trace produces spectral zeta values that access transcendental content beyond the π-polynomial ring of the free operator. (For the *free* Dirac operator D on unit S^3, the SD expansion terminates at exactly two terms by the Jacobi theta modular identity: $\mathrm{Tr}\, e^{-tD^2} \;=\; (\sqrt{\pi}/2)\, t^{-3/2} - (\sqrt{\pi}/4)\, t^{-1/2} + O(e^{-\pi^2/t})$, with $a_k = 0$ for $k \geq 2$; interaction opens the full MZV algebra.)

The Sommerfeld fine-structure sums $D_p = \sum_{n=1}^{\infty} c_p(n)$ (Paper 28, Sec. 7 [12]) are precisely the perturbative expansion of this effect. At each order p in $(Z\alpha)^{2p}$, the coefficients $c_p(n)$ involve generalized harmonic numbers $H_{n-1}^{(r)}$, which are partial Euler sums that access the full Euler–Zagier multiple zeta value (MZV) algebra. The free heat kernel produces only π^{even} (Bernoulli); the Coulomb interaction opens the door to odd-zeta values

($\zeta_R(3)$ at $p = 2$, $\zeta_R(5)$ and $\zeta_R(7)$ at $p = 3$, and so on) and to products of zeta values at higher orders.

This maps directly onto the exchange constant taxonomy:

- **Intrinsic tier** (angular/Gaunt structure): no Mellin transform is needed; the result is purely rational.
- **Calibration tier** (κ, π, $\zeta_R(2)$): Mellin transform of the *free* heat kernel on the compact manifold $\to$ π^{even} exclusively.
- **Embedding tier** (cusp, $1/r_{12}$, Sommerfeld D_p): Mellin transform of the *interaction-deformed* heat kernel $\to$ the full MZV algebra.

c. The product survival rule as a Mellin selection rule. The product survival rule (Paper 28 [12])—the systematic cancellation of $\zeta(2) \times \zeta(\text{odd})$ products at every Sommerfeld order—has a natural interpretation in this framework. The n^2 Fock degeneracy that weights the Sommerfeld sums is the same n^2 that produces $F = \zeta_R(2) = D_{n^2}(4)$ in the free-geometry Mellin output (Paper 2 [7], Phase 4F identification). When the interaction-deformed Mellin transform generates product terms $\zeta(a) \times \zeta(b)$, the Fock-degeneracy weighting acts as a selection rule: products involving $\zeta(2)$ are systematically expelled because $\zeta(2)$ is already "used" as the Fock Dirichlet evaluation F, and the n^2 weighting enforces orthogonality between the free and interacting Mellin outputs.

d. K–Sommerfeld separation as a Mellin consequence. The K–Sommerfeld structural separation (Paper 28 [12])—the PSLQ-verified fact that K/π is not in the $\mathbb{Q}$-span of $\{D_2, \ldots, D_6, 1\}$—is a direct consequence of the Mellin framework. The ingredients of $K = \pi(B + F - \Delta)$ are all *free-geometry* Mellin invariants: B is a truncated trace of the free Casimir, $F = \zeta_R(2)$ is the spectral zeta of the free Fock-degeneracy Dirichlet series at $s = d_{\max}$, and $\Delta^{-1} = g_3^{\text{Dirac}}$ is a single-level mode count of the free Dirac operator. All three live in the π-polynomial ring guaranteed by Eq. (21). The Sommerfeld D_p, by contrast, are Mellin transforms of the *Coulomb-perturbed* heat kernel, dressed with harmonic numbers that access the full MZV algebra. The two sets of quantities inhabit different rings because they are Mellin transforms at different perturbative orders: K at zeroth order (free geometry), D_p at p-th order (interacting geometry).

e. Summary. The Mellin transform (10) is the silent engine of the entire exchange constant taxonomy. It converts discrete spectral data (integer eigenvalues, integer degeneracies) into transcendental content (π, $\zeta(s)$, Dirichlet L-values), and the *class* of transcendental that emerges is determined by three properties of the operator whose heat kernel is being transformed: (1) whether it is first- or second-order (controlling the motivic weight of the Mellin output); (2) whether it acts on scalars or spinors (modulating rational coefficients); and (3) whether the potential is free or interaction-deformed (controlling whether the output stays in the π-polynomial ring or accesses the full MZV algebra). These are precisely the three axes of the QED transcendental classification formalized in Sec. IV.

f. Period-ring stratification of the master Mellin engine (June 2026 closure). The master Mellin engine partition $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ with $k \in \{0, 1, 2\}$ admits a complete period-theoretic stratification, established by three companion sprints in June 2026 (memos `debug/sprint_m1_pure_tate_memo.md`, `debug/sprint_mixed_tate_test_memo.md`, `debug/sprint_m3_cyclotomic_mixed_tate_memo.md`):

$$M_1 \subset \mathbb{Q}[\pi, \pi^{-1}], \qquad M_2 \subset \bigoplus_{k \geq 0} \pi^{2k} \cdot \mathbb{Q}, \qquad M_3 \subset \mathcal{MT}(\mathbb{Z}[i, 1/2]) \text{ at l} \tag{11}$$

indexed by the operator-order slot k. M1 (Hopf-base measure $\mathrm{Vol}(S^2)/4 = \pi$) generates the localised pure-Tate algebra at π; M2 (Seeley–DeWitt heat-kernel coefficients) inherits the Fathizadeh–Marcolli [44] mixed-Tate classification of spectral-action expansions on Robertson–Walker spacetimes, with a pure-Tate refinement on the static S^3 sub-case (no $\zeta(3), \zeta(5)$, or MZV content at the SD level); M3 (vertex-parity sums producing Hurwitz at quarter-integer shifts and Dirichlet L-values including Catalan $G = \beta(2)$ and $\beta(4)$) inherits the Deligne [45] + Glanois [46] cyclotomic mixed-Tate classification at level 4, with the vertex-parity-as-χ_{-4} correspondence realising the Deligne–Glanois Galois descent from level 4 to level 2 on the framework's natural QED observables.

Paper 32 §VIII Corollaries `cor:m1_pure_tate`, `cor:m2_mixed_tate`, and `cor:m3_cyclotomic_mixed_tate` state the case-exhaustion-theorem-level consequences. Every transcendental that has appeared in GeoVac spectral data has a named motivic home indexed by the Mellin slot k; the remainder of this subsection walks the master-mechanism reading (§ master-mechanism), the mechanism-as-domain sharpening (§ domain-partition), the witness panels for the three period-ring closures (§§ Mixed-Tate / M3-cyclotomic / M1-pure-Tate), and the remaining open questions (sectional convention; non-compact rate; discrete c_1).

g. Master-mechanism reading (Track TS-E3, May 2026). The Mellin engine reading admits a sharper structural statement. The spectral-triple framing of GeoVac (Paper 32) admits a candidate case-exhaustion theorem (Paper 32 Theorem on π-source case-exhaustion): every π in a GeoVac observable arises from one of three named mechanisms M1 (Hopf-base measure), M2 (Seeley–DeWitt heat-kernel coefficient), or M3 (vertex-topology Hurwitz Dirichlet L-content). Track TS-E3's falsification sweep against three candidate fourth mechanisms (continuum gravity, anomaly classes, non-trivial principal bundles) returned a stronger structural finding: M1, M2, M3 are not three independent mechanisms but *three sub-cases of a single master* *mechanism* parameterized by operator order:

$$\pi\text{-source} = \mathcal{M}\Big[\mathrm{Tr}(D^k \cdot e^{-tD^2})\Big](s) \quad \text{with} \quad k \in \{0,1,2\}. \tag{12}$$

The three sub-cases correspond to: $k = 0$ (trivial heat kernel on a sphere collapses to the Hopf-base measure $\mathrm{Vol}(S^d)$, recovering M1); $k = 2$ (standard spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ generates Seeley–DeWitt coefficients, recovering M2); and $k = 1$ (Mellin transform of $\mathrm{Tr}(D \cdot e^{-tD^2})$ produces the η-invariant series whose vertex-parity decomposition gives the quarter-integer Hurwitz shifts, recovering M3). The Mellin engine of this section is therefore not just *a* mechanism producing exchange constants; under the case-exhaustion theorem, it is the *master* π-injecting mechanism, with operator order k selecting the sub-mechanism. The first-vs. second-order axis named above (motivic weight) is exactly the $k = 1$ vs. $k = 2$ split; the $k = 0$ free-Mellin case is the Hopf-base measure singled out by Paper 25.

h. Mechanism-as-domain sharpening (Sprint MR-A/B/C, May 2026). The master Mellin engine reading admits a further refinement when read in the *predictive* direction. The formal statement of the sharpening appears as a remark accompanying the case-exhaustion theorem in Paper 32 § VIII [15]; what follows is the operational summary that closes Paper 18's taxonomic engine. A three-track sprint sequence (MR-A: Dirac state-space GH rate; MR-B: spectral-action modular residual; MR-C: L2 next-order constant) tested whether the engine predicts the transcendental ring of a convergence rate from the operator order k of the underlying mechanism.

Sprint MR-B closed the M2 prediction in closed form. The Camporesi–Higuchi heat kernel on the half-integer Dirac spectrum admits a Jacobi ϑ_2 modular decomposition

$$\varepsilon(t) \;=\; \sum_{m\geq 1} (-1)^m \sqrt{\pi}\, e^{-m^2\pi^2/t} \left[t^{-3/2} - 2m^2\pi^2 t^{-5/2} - \tfrac{1}{2} t^{-1/2}\right] \tag{13}$$

verified to > 100 digits at every tested t. Every coefficient sits in the M2 ring $\sqrt{\pi}\cdot\mathbb{Q}\oplus\pi^2\cdot\mathbb{Q}$; the modular exponent is exactly π^2. The leading prefactor $2\pi^{5/2}$, the subleading $-\sqrt{\pi}\, t$, and the $\frac{1}{2}\sqrt{\pi}\, t^2$ correction at the next order are all in M2.

Sprint MR-A computed the natural Dirac analog of the SU(2) scalar central Fejér state-space GH rate. The scalar baseline (L2 sprint, 2026-05-06) had pinned the asymptote rigorously to $\lim_{n\to\infty} n\gamma_n/\log n = 4/\pi$ with explicit uniform bound $\gamma_n \leq 6\,(\log n)/n$ for $n \geq 2$ (rotation-angle normalisation of the SU(2) metric; on the unit S^3 the constant is $2/\pi = \mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2))$, Paper 38, 2026-09-03); the constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2 = 2\,\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2))$ is the Hopf-base measure factor in that normalization, the M1 signature in closed form. The Dirac analog instead obtained the structural identity $\gamma_n^{\mathrm{Dirac}} = \pi$ for every $n_{\max} \geq 1$: the half-integer-only Peter–Weyl truncation cannot span the constant function χ_0, so the kernel does not localize at the identity, and no asymptotic decay exists to PSLQ. The L2 scalar rate decay is in fact an integer×half-integer interference effect; removing the integer-spin sector kills it. This is exactly what the M1 mechanism predicts when the relevant subbundle lacks χ_0, and its presence in the scalar case is precisely what the rigorous $4/\pi$ asymptote records.

Sprint MR-C extracted the L2 next-order constant c in $n \cdot \gamma_n^{\mathrm{scalar}} \sim (4/\pi)\log(n) + c + O(\log(n)/n)$ ($c = 4.1093214674898756204229\ldots$, 22 verified decimal digits; the original 80-digit extraction was float-contaminated from digit 12, see Paper 38's open-questions record), and ran PSLQ at coefficient ceiling 10^4 against M1, M2, M3 and the extended ring $\{\gamma_E, \log 2, G, \zeta(3)\}$. No identification was returned.

The combined verdict reads:

> *Mechanism-as-domain partition.* The index $k \in \{0,1,2\}$ classifies both the M-mechanism and the natural domain of observable from which its signature can be extracted:
>
> - $k = 0$ (M1, Hopf-base measure): state-space GH rates; signature $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ in the SU(2) Fejér asymptote (L2 sprint quantitative-rate theorem).
> - $k = 2$ (M2, Seeley–DeWitt): heat-kernel and spectral-action convergence; signature $\sqrt{\pi}\cdot\mathbb{Q}\oplus\pi^2\cdot\mathbb{Q}$ throughout the modular residual (13).
> - $k = 1$ (M3, vertex-topology Hurwitz): vertex-restricted parity-character spectral sums; signature Dirichlet $\beta(s)$ content via the closed-form identity $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ (Paper 28 Theorem 3 [12]); witnessed in Paper 28 § QED-vertex (Catalan G, $\beta(4)$) but *not* in state-space GH rates.

Mixing the index of the question (e.g. "state-space GH rate") with the index of the mechanism ("vertex parity") produces structural degeneracy: this is exactly what MR-A measured. MR-C's negative is consistent with the same reading: the L2 subleading constant is downstream of an M1 Stein–Weiss IBP, not a direct $\mathcal{M}[\mathrm{Tr}(D^k \cdot e^{-tD^2})]$, and the predictive engine does not commit to the rings of derivative quantities along subsequent analytical pipelines. The engine is predictive *within* each domain, where the domain is determined by k. Source memos: `debug/mr_{a,b,c}_*_memo.md`, data `debug/data/mr_{a,b,c}_*.json`.

Sprint Q5'-Stage1-2b (June 2026, `debug/sprint_q5p_2b_phi0_mellin_memo.md`) sharpens the M_1 sub-mechanism's π injection further: on a compact 3-dimensional spectral triple, the Hopf-base measure π enters the formal Mellin transform $\Gamma(s)\cdot\zeta_{D^2}(s)$ specifically at the spectral-dimension pole $s = d/2 = 3/2$ via the Weyl-law short-time leading $\mathrm{Tr}(e^{-tD^2}) \sim \sqrt{\pi}/2 \cdot t^{-d/2}$; at every finite truncation $n_{\max}$ the spectral zeta itself is rational-valued, and the master-Mellin transcendentals appear only in the continuum analytic-continuation step. This localizes the M_1 sub-mechanism's natural domain to the non-integer spectral-dimension residue of the cochain-level Mellin transform. The same-day continuum follow-on (Sprint Q5'-Stage1-2b-continuum, `debug/sprint_q5p_continuum_mellin_memo.md`) closes the M_1 identification at theorem grade: the continuum spectral zeta $\zeta^{\rm cont}_{D^2}(s) = 2\,\zeta(2s-2, 3/2) - \frac{1}{2}\,\zeta(2s, 3/2)$ has a simple pole at $s = 3/2$ with bit-exact meromorphic residue 1, so $\Gamma(s)\,\zeta^{\rm cont}_{D^2}(s)$ has Mellin residue $\Gamma(d/2) = \sqrt{\pi}/2$ at $s = d/2$. This coefficient is the same spectral object as the L2 state-space GH asymptote $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ of Paper 38 § VIII and as the Hopf-base Haar normalization $\mathrm{Vol}(S^2)/4 = \pi$ of § III.2, related by exact factors $\pi^{3/2}/8$ and $\pi^2/4$ respectively with no spectral-data input. The M2 continuum panel at integer $s \in \{1, 2, 3, 4, 5\}$ reproduces the Sprint Q5'-CH-2 Seeley–DeWitt panel bit-exactly. Tauberian rate-uniformity in a neighborhood of $s = 3/2$ is the Stage-2-relevant named open gap, reachable from Karamata [49] and Korevaar [50].

i. Connection to motivic / period theory (Sprint Mixed-Tate Test, June 2026). The master Mellin engine reading places the GeoVac taxonomy in direct contact with the published period / motivic-Galois program for spectral-action heat-kernel coefficients on continuum substrates. Fathizadeh–Marcolli [44] prove that the coefficients of the asymptotic expansion of the Connes–Chamseddine spectral action on Robertson–Walker spacetimes $\mathbb{R}\times S^3$ are periods of mixed Tate motives over $\mathbb{Q}$. Rejzner's monograph-scale treatment [47] develops the parallel period-theoretic framework inside perturbative Algebraic Quantum Field Theory. Both lines descend from the Connes–Marcolli [48] renormalization / motivic-Galois program and the Marcolli–van Suijlekom gauge-network spectral-triple framework [51] (with the Perez-Sanchez correction [52, 53] that the continuum limit is Yang–Mills without Higgs), of which GeoVac is an instance at the data level and an extension at the structural level. Brown's mixed-Tate-motives-over-$\mathbb{Z}$ theorem [54] provides the period-ring backbone underlying the M3 cyclotomic-mixed-Tate classification. Sprint Mixed-Tate Test (June 2026, memo `debug/sprint_mixed_tate_test_memo.md`) closes the natural inheritance question for the GeoVac S^3 sub-case in the POSITIVE direction with a pure-Tate refinement: at the standard volume-normalized convention, the GeoVac discrete M2 Seeley–DeWitt coefficients on S^3 sit inside the strictly smaller sub-ring $\bigoplus_k \pi^{2k}\cdot\mathbb{Q}$ of even-weight pure Tate periods, with no $\zeta(3)$, $\zeta(5)$, or MZV content appearing at the SD level. This is sharper than the F–M generic R–W case in two ways: (i) the GeoVac S^3 Dirac SD expansion is two-term exact ($a_k = 0$ for $k \geq 2$, Paper 51 Cor. 2.1), not generically infinite; (ii) the scalar Laplacian SD coefficients have the closed-form geometric-series collapse $a_k^{\Delta} = 2\pi^2/k!$, not the generic R–W expansion. The M3 sub-mechanism classification (companion Sprint M3 Cyclotomic Mixed-Tate, June 2026, memo `debug/sprint_m3_cyclotomic_mixed_tate_memo.md`) places M3 in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4 at depth equal to loop order, by direct application of Deligne [45] and the explicit basis of Glanois [46]: un-restricted Dirac Dirichlet $D(s)$ lies in $\mathcal{MT}(\mathbb{Z})$ at depth 1; vertex-restricted $D_{\rm even} - D_{\rm odd}$ lies in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level 4 depth 1 via quarter-integer Hurwitz + Dirichlet β (Catalan G, $\beta(4)$); the depth-k loop tower lives at depth k in $\mathcal{MT}$ at level ≤ 4. M3 does *not* contaminate the SD coefficients (M3 lives at $k = 1$ in $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$, M2 at $k = 2$). Together with the companion Paper 32 § VIII corollaries `cor:m2_mixed_tate` and `cor:m3_cyclotomic_mixed_tate`, the master Mellin engine now has a complete period-theoretic stratification: M1 in $\mathbb{Q}[\pi, 1/\pi]$, M2 in $\bigoplus_k \pi^{2k}\cdot\mathbb{Q}$, M3 in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4.

j. M3 sectional convention refinement (Sprint L2-D, May 2026). The vertex-parity Hurwitz mechanism of M3 is sectional in the chosen index parity convention, NOT in spacetime signature. Specifically: at the Krein extension to $(3, 1)$ (Sprint L2-D, Paper 32 § VIII.D [15]), M3's L0 trivialization prediction holds trivially under chirality-pairing (chirality symmetry makes $D_{\rm even} = D_{\rm odd}$ identically) but falsifies under Paper 28's $n_{\rm Fock}$-parity reading — the convention that accesses Catalan $G = \beta(2)$ via the closed-form identity $D_{\rm even}(s) - D_{\rm odd}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ (Paper 28 Theorem 3 [12]). The Bizi–Brouder–Besnard sign-flip $\{J, \gamma^5\} = 0$ at $(m, n) = (4, 6)$ is a spectral-triple-axiom statement, not a vertex-parity-sum statement. M3's transcendental content ($\beta(2)$, $\beta(4)$, Catalan G, Dirichlet L-values) is therefore tied to the $n_{\rm Fock}$-parity sector choice of the truncation, not to the signature of the underlying spectral triple. This refinement does not change the master Mellin engine domain partition above (M1 $\leftrightarrow$ state-space GH rates, M2 $\leftrightarrow$ heat-kernel and spectral-action convergence, M3 $\leftrightarrow$ vertex-restricted parity-character spectral sums); it sharpens the parity-convention-dependence of the M3 sub-mechanism's accessibility. Source memos: `debug/l2_d_connes_axiom_audit_31_memo.md`, data `debug/data/l2_d_connes_axiom_audit_31.json`.

k. Operator-vs.-slot refinement of the M3 sub-mechanism (Sprint NA-1, June 2026). The Mellin-slot index $k \in \{0, 1, 2\}$ in $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ classifies the *sub-mechanism* (M1 / M3 / M2); it does *not* by itself fix the period ring of the output. The Mellin slot exponent s — the argument at which the moment is evaluated — selects *which periods inside the sub-mechanism's natural period ring* are accessed. Sprint NA-1 (`debug/sprint_na1_depth2_mellin_memo.md`, 2026-06-06) makes this distinction explicit on the natural Camporesi–Higuchi diagonal substrate for the $k = 1$ slot

(M3). Let

$$M_3^{\gamma_P}(r) \;:=\; \sum_{n\geq 0}(-1)^n\, g_n\, |\lambda_n|^{-r} \qquad (14)$$

be the parity-graded Mellin function of the Camporesi–Higuchi Dirac spectrum, with $\gamma_P|n\rangle = (-1)^n|n\rangle$ the vertex-parity grading on the eigenshells. The *same operator* γ_P produces two structurally distinct period-ring outputs depending on the Mellin slot exponent at which it is probed:

- **Pure-Tate slot** ($r = 2s-1 \in 2\mathbb{Z}+1$, odd integer; Paper 55 [17] § M3-diagonal-collapse Theorem on $M_3^{\gamma_P}$ closed form): closed form $M_3^{\gamma_P}(s) = 2^{2s-3}\,(\beta(2s-1)-\beta(2s-3))$, landing in $\bigoplus_k \pi^{2k+1}\cdot\mathbb{Q}$ via the Euler-number closed form $\beta(2k+1) \in \pi^{2k+1}\cdot\mathbb{Q}$. This is *pure-Tate at level* 1.

- **Level-4 cyclotomic slot** (integer s on the unshifted Dirac Dirichlet difference; Paper 28 Theorem 3 [12]): $D_{\mathrm{even}}(s)-D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s)-\beta(s-2))$, accessing β(even) values (Catalan $G = \beta(2)$, $\beta(4)$) inside $\mathcal{MT}(\mathbb{Z}[i,1/2])$ at level 4 (Paper 55 [17] § M3-Galois-descent Proposition on vertex parity as Deligne–Glanois descent).

Both probes act on the same diagonal parity grading γ_P sitting in the $k = 1$ slot of the master Mellin engine; the period content (pure-Tate vs. level-4 cyclotomic) is selected by the Mellin slot exponent r, not by a different operator. The Glanois [46] basis $\mathcal{B}^4$ accommodates both columns: the pure-Tate column is the $N = 1$ sub-basis, the level-4 column is the genuinely $N = 4$ content.

This refinement does not change the case-exhaustion statement (Paper 32 § VIII), the M3 sectional convention refinement above, or the period-ring stratification $M_3 \subset \mathcal{MT}(\mathbb{Z}[i,1/2])$ at level ≤ 4. It sharpens the description of the M3 row of the master Mellin engine: the sub-mechanism index $k = 1$ commits to the η-invariant Mellin family $\mathcal{M}[\mathrm{Tr}(D\cdot e^{-tD^2})]$, but the period-ring location within M_3 at each evaluation is determined by the slot exponent. Cross-reference Paper 55 [17] § M3-diagonal-collapse and § M3-Galois-descent for the explicit pure-Tate vs. level-4 demonstration, including the NA-1 diagonal-collapse identity $J(s_1,s_2) = M_3^{\gamma_P}(s_1+s_2-1)$ on the Camporesi–Higuchi diagonal substrate.

An identical operator-vs.-slot reading is available for M2 (the Seeley–DeWitt slot, $k = 2$): the squared-Dirac heat-kernel operator D^2 is fixed, but the Mellin slot exponent s selects which Seeley–DeWitt coefficient $a_k(D^2)$ is read off via $\mathrm{Res}_{s=(d-2k)/2}\,\zeta_{D^2}(s)$. The pure-Tate M2 closure $M_2 \subset \bigoplus_k \pi^{2k}\cdot\mathbb{Q}$ (Sprint Mixed-Tate Test, paragraph above) describes the period ring; which $\pi^{2k}\cdot\mathbb{Q}$ slot is accessed by a given observable is fixed by the Mellin slot exponent at which the heat-kernel trace is evaluated. The M1 slot $k = 0$ is degenerate in this sense: the trivial heat kernel collapses to the Hopf-base measure $\mathrm{Vol}(S^d)$, and the period ring $\mathbb{Q}[\pi,\pi^{-1}]$ has no internal Mellin-slot stratification to select. The operator-vs.-slot distinction is therefore non-trivial for M2 and M3 (whose period rings have internal weight gradings) and degenerate for M1 (whose period ring has only one graded piece, $\mathbb{Q}[\pi,\pi^{-1}]$ at depth 0).

l. Why slot, not operator (Sprint NA-1-offdiag, June 2026). The structural reason the operator-vs.-slot distinction works as stated above: the trace functional $\mathrm{Tr}(\cdots e^{-t_iD^2}\,\gamma\, e^{-t_jD^2}\cdots)$ commits to the *diagonal* of any inserted operator γ in the D-eigenbasis and loses the off-diagonal information. For any self-adjoint γ on a finite-dimensional Hilbert space — commuting or non-commuting with D —

$$\mathrm{Tr}\big(D^2\, e^{-t_1D^2}\,\gamma\, D\, e^{-t_2D^2}\big) \;=\; \sum_i \widetilde{\gamma}_{ii}\,\lambda_i^3\, e^{-(t_1+t_2)\lambda_i^2},$$

depending only on t_1+t_2 (Paper 55 [17] § M3-diagonal-collapse, Theorem on the off-diagonal CH substrate). The substrate change from a diagonal γ-grading to a non-diagonal one (e.g. chirality-flipping E_1 entries on the full spinor bundle) does not change the trace output, because off-diagonal entries of $\widetilde{\gamma}$ are invisible to the trace. This is the deeper structural reason the operator-vs.-slot reading commits to the slot exponent: the operator side of the master Mellin engine is projected onto its diagonal in the spectral basis before any period content is read off, and only the slot exponent s then selects which period ring element is accessed.

m. Compact / non-compact partition of the $4/\pi$ rate (Sprint M-Z, May 2026). The master Mellin engine's M_1 Hopf-base measure mechanism produces the $4/\pi$ rate constant (rotation-angle normalisation; $2/\pi$ on the unit S^3) in the state-space Gromov–Hausdorff convergence of Paper 38 (compact $SU(2)$ truncation) and in the universal rate theorem of Paper 40 (compact connected semisimple Lie groups with bi-invariant metric; a circle factor carries $2/\pi$). Sub-sprint M-Z (modular propinquity investigation, 2026-05-23) flagged a structural reason why this rate does *not* transport to non-compact Coulomb (Sturmian basis on $L^2(\mathbb{R}^3)$): the Hopf-base measure mechanism is compact-Lie-quotient specific; Sturmian's continuum content dilutes the Plancherel weight to 1, killing the localization that drives the $\log n/n$ rate on $SU(2)$. A rigorous non-compact analog of the Paper 38 / Paper 40 rate remains an open question; Latrémolière 2018 [43] establishes existence of a complete propinquity metric on the inductive-limit construction but not a rate. Source: `debug/sprint_modular_propinquity_mZ_bethe_log_memo.md`.

n. M1 period-ring closure (Sprint M1 Pure-Tate, June 2026). The M1 sub-mechanism produces period values in the localised pure-Tate sub-ring $\mathbb{Q}[\pi,\pi^{-1}]$ of mixed Tate periods over $\mathbb{Q}$, at depth 0 and Tate weight $\in\mathbb{Z}$, by direct identification of $\pi = \mathrm{Vol}(S^2)/4$ as a weight-1 Tate period (Kontsevich–Zagier 2001). Every M1 witness logged across the corpus — $\mathrm{Vol}(S^2)/4 = \pi$ (Paper 25 Hopf bundle base measure); $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ (Paper 38 §L2 asymptotic state-space GH rate); $4/\pi$ (Paper 40 Thm 1 universal rate constant across compact connected semisimple Lie groups); $1/\pi^2$ (Paper 43 §10.2 Pythagorean orthogonality); $1/\pi^2$ inside $-3\zeta(3)/(16\pi^2)$ (Paper 50 §3 F-theorem, an M1×M3 cross-product) — is a $\mathbb{Q}$-linear combination of π^n for $n \in \mathbb{Z}$. Combined with this sprint's two companion closures (Sprint Mixed-Tate Test placing M2 in $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, and Sprint M3 Cyclotomic Mixed-Tate placing M3 in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4), the master Mellin engine of § III.7 has a complete period-theoretic stratification indexed by the Mellin slot $k \in \{0, 1, 2\}$:

$$M_1 \subset \mathbb{Q}[\pi, \pi^{-1}], \quad M_2 \subset \bigoplus_k \pi^{2k} \cdot \mathbb{Q}, \quad M_3 \subset \mathcal{MT}(\mathbb{Z}[i, 1/2]) \text{ at level} \leq 4.$$

Companion Paper 32 §VIII Corollary `cor:m1_pure_tate` states the case-exhaustion-theorem-level consequence; memo `debug/sprint_m1_pure_tate_memo.md`.

o. Discrete c_1 verification (Sprint TS-E3 falsification target). The strongest concrete computational falsification target flagged by Track TS-E3 is the discrete first Chern number of the Hopf U(1) bundle on the finite Fock-projected S^3 graph: under the master Mellin engine reading, the discrete c_1 should be *integer-valued* on any finite cutoff, because the construction engages no continuous integration step (no Hopf-base measure, no heat-kernel trace, no Mellin transform), and therefore none of M1, M2, M3. Both physically natural $U(1)$ phase conventions (Condon–Shortley canonical, all phases zero; and Z_2 alternating, $\phi = \pi$ on $L_\pm$ edges) yield exactly $c_1 = 0$ at $n_{\max} = 2$ and $n_{\max} = 3$, confirmed in exact rational arithmetic (Fraction-typed phases, no floating-point ambiguity). A sanity-check convention (prescribed flux 1/4 per plaquette via non-tree-edge phases) recovers $|c_1| = (\#\text{cycles})/4 = 1/2$ at $n_{\max} = 3$ exactly, and a stress-test convention (1/3 flux) recovers $|c_1| = 2/3$, confirming the algorithm tracks rational non-integer fluxes faithfully and would surface a non-integer result if one existed under the natural conventions. The continuum first Chern integer reappears via the M1 $\mathrm{Vol}(S^2)/4 = \pi$ normalisation in the continuum limit; on the finite graph the topological invariant is integer in $\mathbb{Q}$. Source: `debug/discrete_c1_hopf_s3.py`, data `debug/data/discrete_c1_hopf_s3.json`, memo `debug/discrete_c1_falsification_memo.md` (Sprint TS-E3, 2026-05-04).

IV. TAXONOMY OF EXCHANGE CONSTANTS

The following taxonomy classifies exchange constants by the type of compactness mismatch they mediate. The hierarchy—intrinsic, conformal/calibration, embedding, algebraic-implicit, composition, and inner-factor input data—can be read as increasing departure from compactness: intrinsic constants convert between compact manifolds of different dimension; conformal/calibration constants convert between a compact manifold and flat space; embedding constants arise from coordinate singularities in the non-compact description; algebraic-implicit functions encode parameter-dependent geometry through a characteristic polynomial whose coefficients are inherited from upstream constants; composition constants measure the cost of factorizing a single compact space into multiple incompatible coordinate systems; and inner-factor input data are the calibration constants of the almost-commutative inner factor (the Standard-Model Yukawa/Higgs sector), supplied externally rather than generated by any projection.

Based on the catalog and the Weyl–Selberg identification, we propose the following taxonomy:

TABLE I. Spectral-geometric exchange constants in the GeoVac hierarchy. "Source" indicates the discrete structure; "Target" indicates the continuous coordinate system. "Type" classifies the exchange constant by what determines it. "Algebraic (implicit)" indicates that the function satisfies a polynomial equation $P(R, \mu) = 0$ with coefficients inherited from upstream exchange constants (Sec. VII E). "Piecewise" indicates that the angular Hamiltonian has piecewise-smooth parameter dependence (from the split-region Legendre expansion), preventing a global characteristic polynomial. The function is piecewise-algebraic within each region.

Level	Source	Target	Constant	
—	S^1 lattice	$\mathbb{R}^1$	π	
1	S^3 graph	$\mathbb{R}^3$	$\kappa = -1/16$	
2 (m=0)	Laguerre rec.	prolate sph.	(none)	
2 (m≠0)	assoc. Laguerre	prolate sph.	$e^a E_1(a)$	
3 (FCI)	$S^3 \times S^3$	(none)	(none)	
3 (hyper.)	SO(6) reps	$S^5(R)$ param.	$\mu(R)$	a
4 (prolate CI)	prolate product	(none)	(none)	
4 (hyper.)	SO(6) reps	(ρ, R) param.	$\mu(\rho, R)$	
5 (composed)	Level 3+4 fibers	full N-electron	PK pseudopot.	
5 (balanced)	hydrogenic × multipole	cross-center V_{ne}	$\gamma(k, \alpha R)$	

The taxonomy reveals a hierarchy of exchange types:

1. **Algebraic** (no exchange constant): When the computation stays on the graph or uses a basis whose inner product matches the graph structure, no transcendental appears. Examples: Level 1 on S^3, Level 2 σ states, Level 3 FCI.

2. **Intrinsic constants** ($\pi^{d/2}$): Determined by the manifold's dimension and volume alone, independent of any discretization or basis choice. These are the Weyl constants—universal volume-to-eigenvalue conversion factors. Example: π itself, for $S^1 \to \mathbb{R}$.

3. **Conformal constants**: Determined by the conformal geometry of the projection from the compact graph manifold to flat space. Example: $\kappa = -1/16 = -1/\Omega^4(0)$, where $\Omega(0) = 2$ is the stereographic conformal factor at $p = 0$ (Sec. VII F). Equivalently derivable from graph vertex degree $d_{\max} = 4$; both routes yield the same value because the graph discretizes S^3.

4. **Embedding constants**: Single transcendental numbers seeding algebraic recurrences. Determined by the interaction between a spectral basis and a coordinate singularity. Examples: $e^a E_1(a)$ for the prolate centrifugal pole; the $1/r_{12}$ electron-electron cusp, which is the embedding exchange constant for the two-electron coalescence in any product basis. The cusp embedding constant is unique in being *removable* by a similarity transformation: the transcorrelated (TC) Jastrow factor $J = \frac{1}{2}r_{12}$ satisfies $e^{-J}(1/r_{12})\,e^J$ = smooth, converting the algebraic partial-wave convergence $1/(l+\frac{1}{2})^4$ to exponential. This makes $1/r_{12}$ a "soft" embedding constant—present in any product-space representation but eliminable by a non-unitary change of basis—in contrast to $e^a E_1(a)$, which is structurally irreducible.

   The $1/r_{12}$ classification is *universal*: the cusp condition survives relativistic corrections with only an $O((Z\alpha)^2)$ amplitude rescaling (Kutzelnigg-Morgan 1992), and partial-wave convergence exponents (l^{-4} singlet, l^{-6} triplet) are identical in Schrödinger-Coulomb and Dirac-Coulomb (verified numerically at $Z = 4$). The Dirac program does not regularize the e-e cusp; full QED would be required.

   A *distributional* sub-category appears in the Breit spin-spin interaction: the bare $1/r_{12}^3$ kernel diverges logarithmically at coalescence for every partial wave. The Breit-Pauli tensor projection regularizes this, leaving $K_l^{BP} = r_<^l/r_>^{l+3}$ with exact rational matrix elements for same-n orbitals ($R^2_{BP}(2p,2p;2p,2p) = 7/768$ at $Z = 1$). For cross-n orbitals a new transcendental appears: $M^2_{\rm ret}(1s,2p;1s,2p) = 21344/729 - (10240/243)\log(2)$, the first log-embedding constant in the framework.

   On the S^3 pair-state lattice (v2.9.1), 92–94% of V_{ee}'s Frobenius mass is diagonal in the graph eigenbasis, with the off-diagonal residual concentrated on the $(1s,1s)$ coalescence pair-state. The graph nearly diagonalizes V_{ee} without containing $1/r_{12}$—direct evidence that the cusp is embedding content the graph cannot absorb, explaining the graph-native CI convergence floor (Paper 13, Track DI).

5. **Algebraic (implicit) functions**: Functions of continuous parameters defined implicitly by $P(R,\mu) = 0$, where P is the characteristic polynomial of the parameter-dependent Hamiltonian $H(R) = H_0 + R \cdot V_C$. The coefficients of P lie in $\mathbb{Q}(\pi,\sqrt{2})$—a transcendental extension of $\mathbb{Q}$—but the R-dependence itself is algebraic: $\mu(R)$ is a root of a polynomial in R. The transcendental content is *inherited* from the angular coupling matrix V_C (which contains upstream exchange constants of embedding type), not generated by the parameterization. Example: $\mu(R)$ for the SO(6) $\to$ SO(4) symmetry crossover on S^5 at Level 3. At $l_{\max} = 0$, $\mu(R)$ is an explicit linear function of R; at higher $l_{\max}$, it is an algebraic function of degree $l_{\max}+1$, with branch points at eigenvalue crossings in the complex R-plane. At Level 4, the split-region Legendre expansion of the nuclear charge function introduces piecewise-smooth ρ-dependence at the region boundary $\rho = 1/2$, breaking the linear pencil structure. The angular eigenvalues $\mu(\rho)$ are piecewise-algebraic within each region but do not satisfy a single global characteristic polynomial. The product-space CI saturation at 92.5% D_e remains the key irreducibility result.

6. **Composition constants**: The accuracy cost of factorizing a full N-electron wavefunction into composed coordinates. Unlike the preceding types, which convert between discrete and continuous descriptions, composition constants convert between a single-geometry and a multi-geometry (factorized) description. The factorization destroys inter-group antisymmetry—the exchange operator P_{ij} mixes coordinate systems that composition separates—and the PK pseudopotential is the local approximation to this nonlocal effect. Example: PK for LiH composed geometry (Level 3 core + Level 4 valence), which achieves 144× angular compression at the cost of ∼5% $R_{\rm eq}$ error at $l_{\max} = 2$.

7. **Inner-factor input data**: The calibration constants of the almost-commutative inner factor—parameter-tied Dirichlet rings generated by the eigenvalues of a finite Dirac operator D_F on a Connes–Chamseddine-compatible inner factor (the Standard-Model Yukawa/Higgs sector). Unlike the preceding types, which a projection *generates* from the discrete graph, inner-factor data are empirical input from outside the framework: GeoVac admits a Higgs sector structurally but does not autonomously select the Yukawa couplings (Sprint H1; Paper 32 § VIII.C), so these constants are inputs, not outputs. They are categorically disjoint from the outer-factor period content (the $M_1/M_2/M_3$ Mellin sectors) and tensor cleanly with it, neither determining the other.

The dependency hierarchy is explicit: intrinsic constants depend only on dimension; conformal/calibration constants depend on the graph; embedding constants depend on the basis; algebraic-implicit functions depend on the full parameterized geometry; composition constants depend on the factorization topology (which groups share coordinates); and inner-factor input data depend on the choice of finite spectral triple, supplied from outside the framework.

a. Odd-zeta content from first-order spectral operators. The Dirac operator on unit S^3 (Camporesi–Higuchi [31] spectrum $|\lambda_n| = n+3/2$, degeneracy $g_n^{\rm Dirac} = 2(n+1)(n+2)$) introduces a structurally new transcendental: odd-zeta values $\zeta_R(2k+1)$ enter the Dirac Dirichlet series $D(s) = \sum_n g_n^{\rm Dirac}/|\lambda_n|^s$ at odd integer $s \geq 5$.

The Hurwitz decomposition (Theorem 1) gives

$$D(5) \;=\; 14\,\zeta_R(3) \;-\; \tfrac{31}{2}\,\zeta_R(5), \tag{15}$$

the first integer s at which $\zeta_R(3)$ appears with a non-divergent coefficient (at $s=3$ the term proportional to $\zeta_R(s{-}2) = \zeta_R(1)$ diverges). At the packing exponent $s=4$, by contrast,

$$D(4) \;=\; 6\,\zeta_R(2) - \tfrac{15}{2}\,\zeta_R(4) \;=\; \pi^2 - \frac{\pi^4}{12}, \tag{16}$$

purely π^{even}—consistent with Theorem 1 below (squaring the spectrum maps every odd-power Dirichlet sum to an even power, and $\zeta_{D^2}(s)$ lies in $\sqrt{\pi}\cdot\mathbb{Q} \oplus \pi^2\cdot\mathbb{Q}$ at integer s). The half-integer eigenvalue $|\lambda_n| = n+3/2$ is essential: naive substitution of an integer m for the eigenvalue would predict an even-s odd-zeta witness, contradicting Theorem 1 part (2) and Paper 28 [12] Theorem 5 / Eq. (43). The shift $3/2$ is what cancels the singular $m=1$ term in the odd-integer rewrite of the spectral sum.

This odd-zeta content (entering at odd s on the spinor bundle) is categorically different from calibration-tier content. Calibration constants carry even-zeta or π^{even} content arising from second-order Riemannian operators through Jacobi-θ inversion; the odd-zeta $\zeta_R(3)$ enters through the first-order Dirac operator via half-integer Hurwitz-zeta identities at *odd s*, and is $\mathbb{Q}$-linearly independent of $\zeta_R(2)$ by Apéry's theorem. Within Paper 2's $K = \pi(B+F-\Delta)$, the odd-zeta tier is a byproduct rather than a summand: $\zeta_R(3)$ is structurally present on S^3 without participating in K.

b. Spinor-intrinsic content from first-order operators on a spinor bundle. The Breit–Pauli reduction of the Dirac equation introduces a second new transcendental tier: α^2 and $\gamma := \sqrt{1-(Z\alpha)^2}$. The leading spin-orbit coupling

$$H_{\text{SO}} \;=\; \tfrac{Z\alpha^2}{2}\,\tfrac{1}{r^3}\,\mathbf{L}\cdot\mathbf{S}, \tag{17}$$

evaluated in the (κ, m_j) basis via Szmytkowski's reduction $\langle \mathbf{L}\cdot\mathbf{S}\rangle = -(\kappa+1)/2$ and the hydrogenic $\langle 1/r^3\rangle_{n,l} = Z^3/[n^3\,l(l+\tfrac{1}{2})(l+1)]$, gives

$$\langle n,\kappa,m_j|\,H_{\text{SO}}\,|n,\kappa,m_j\rangle \;=\; -\,\frac{Z^4\,\alpha^2\,(\kappa+1)}{4\,n^3\,l(l+\tfrac{1}{2})(l+1)}, \tag{18}$$

with Kramers cancellation at $l=0$. Full-Dirac radial corrections enter through γ, which satisfies

$$\gamma^2 + (Z\alpha)^2 \;=\; 1, \tag{19}$$

making γ algebraic of degree 2 over $\mathbb{Q}(\alpha^2)$. Together α^2 and γ generate the ring

$$R_{\text{sp}} \;:=\; \mathbb{Q}\big(\alpha^2\big)\big[\gamma\big]/\big(\gamma^2+(Z\alpha)^2-1\big), \tag{20}$$

in which every spinor-intrinsic coefficient lies—mechanically certified by `geovac/spinor_certificate.py`.

This tier is categorically distinct from both the odd-zeta tier and calibration: α^2 is a *coupling* content appearing in Hamiltonian coefficients (not in the basis or spectrum), depending only on the existence of spin coupling in the operator. Higher-order corrections (Darwin α^4, mass-velocity α^4, Breit α^2 in V_{ee}) populate the same ring R_{sp} without creating new transcendental tiers.

c. Cross-reference to Paper 34's projection taxonomy. In the projection-dictionary language of Paper 34 § III, the spinor-intrinsic ring R_{sp} is the transcendental signature of the *Camporesi–Higuchi spinor-lift projection* (Paper 34 § III.7), and that projection is the unique source of half-integer Hurwitz / odd-ζ / Dirichlet-β content in the fifteen-projection dictionary. Sprint TX-A (May 2026, Paper 34 § VII Theorem 3 / composition table) confirmed this assignment as exclusive: no other projection in the catalogue introduces odd-zeta content into the transcendental signature, and the spinor-lift projection is the lone bridge between the calibration-tier (π^{even}) world of second-order Riemannian operators and the odd-zeta world of first-order Dirac operators. Equivalently: seeing $\zeta_R(3)$, Catalan's G, or any Dirichlet-β value in a GeoVac observable is diagnostic of the spinor-lift projection having been invoked at some step in the chain.

TABLE II. Transcendental content classified by operator order and bundle type. The (2nd-order, spinor) cell is degenerate with scalar calibration (π^{even} only)—see the D^2 discussion below.

Tier	Example transcendentals	Operator order	Bun
Intrinsic	$\pi^{d/2}$ Weyl prefactors	—	scal
Calibration	$\kappa=-1/16$, $F=\pi^2/6$	2nd (Δ_{LB})	scal
Embedding	$e^a E_1(a)$, $1/r_{12}$	—	grap
Flow / composition	$\mu(\rho,R)$, PK	—	mul
Odd-zeta	$\zeta_R(3), \zeta_R(5), \ldots$	1st (D)	Dira
Spinor-intrinsic	α^2, γ	1st (D / Breit–Pauli)	spin
D^2 calibration	π^{even} (degenerate)	2nd (D^2)	spin

d. Operator-order × bundle classification. Tier 1 (ζ) enters through the *spectrum* of the Dirac operator; Tier 2 (α^2, γ) enters through the *coupling* of spin to orbital angular momentum in the Hamiltonian coefficients. Neither introduces π beyond the calibration tier.

e. Operator-order as transcendental discriminant. The squared Dirac operator D^2 on unit S^3 fills the last open cell of Table II. We promote the discriminating observation to a theorem.

Theorem 1 (Operator-order discriminant on S^3)**.** *Let D be the Camporesi–Higuchi Dirac operator on unit S^3 with spectrum $|\lambda_n| = n+3/2$ and degeneracy $g_n^{\text{Dirac}} = 2(n+1)(n+2)$. Then:*

1. *The spectral zeta of the squared operator D^2 admits the exact closed form*

$$\zeta_{D^2}(s) \;=\; 2^{2s-1}\big[\lambda(2s{-}2) - \lambda(2s)\big], \qquad s\in\mathbb{Z}_{\geq 2}, \tag{21}$$

where $\lambda(a) = (1-2^{-a})\,\zeta_R(a)$ is the Dirichlet lambda function. Every value lies in $\mathbb{Q}[\pi^2]$ (π^{even} content exclusively).

2. *The first-order spectral zeta $D(s) = \sum_n g_n^{\text{Dirac}} |\lambda_n|^{-s}$ admits the Hurwitz decomposition*

$$D(s) \;=\; 2\big(2^{s-2}-1\big)\,\zeta_R(s-2) - \tfrac{2^s-1}{2}\,\zeta_R(s), \qquad s \in \mathbb{Z}_{\geq 2}. \tag{22}$$

At even s both terms are π^{even}; at odd $s \geq 3$ the terms are $\mathbb{Q}$-linearly independent of π (Apéry 1978 for $\zeta_R(3)$; Goncharov–Zagier framework for higher odd zetas). In particular $D(4) = 6\zeta_R(2) - \frac{15}{2}\zeta_R(4) = \pi^2 - \pi^4/12$ is π^{even}, while the first odd-zeta witness is $D(5) = 14\,\zeta_R(3) - \frac{31}{2}\zeta_R(5)$, carrying odd-zeta content not present in any value of ζ_{D^2}.

Operator order—first vs. second—is therefore the primary transcendental discriminant on round S^3. Bundle type modulates degeneracies, eigenvalue lattices, and rational coefficients, but not the transcendental class.

Proof sketch. For (1): Lichnerowicz [30] gives $D^2 = \nabla^*\nabla + R/4$ with scalar curvature $R/4 = 3/2$, so D^2 has a rational shift relative to the conformal Laplacian. The Camporesi–Higuchi spectrum yields $|\lambda_n|^2 = (n+3/2)^2$, which after the half-shift identity for the Hurwitz zeta reduces to Eq. (21). Bernoulli's formula $\zeta_R(2k) = \text{rational} \times \pi^{2k}$ then closes the case. For (2): regrouping the same Camporesi–Higuchi sum without squaring keeps the half-integer shift, producing the Hurwitz combination of Eq. (22); squaring maps odd-power spectral sums to even-power sums, structurally eliminating the $\zeta_R(2k{+}1)$ visible in the unsquared sum and reducing the (2nd-order, spinor) cell to scalar calibration. Computational verification: `tests/test_qed_vacuum_polarization.py` checks Eq. (21) symbolically and against a direct Camporesi–Higuchi spectral sum, and `tests/test_qed_two_loop.py` verifies the odd-zeta content of Eq. (22) via the Hurwitz closed form and a PSLQ parity classification. □

f. Three-tier collapse (corollary). As a corollary of Theorem 1, the operator-order × bundle taxonomy of Table II collapses to three effective tiers:

1. **First-order operators:** odd-zeta content ($\zeta_R(3), \zeta_R(5), \ldots$) plus spinor coupling content (α^2, γ).
2. **Second-order operators:** π^{even} exclusively, on both scalar and spinor bundles.
3. **Basis and composition tiers:** embedding, flow, composition constants (operator-independent).

This is consistent with Paper 24's Coulomb/HO asymmetry [9]: second-order Riemannian operators introduce π through nonlinear conformal projections, while first-order operators (the HO Bargmann Euler operator on $H^2(S^5)$, the Dirac operator on S^3) have π-free spectra in rational arithmetic.

g. Motivic weight and operator order. The operator-order discriminant—second-order operators produce π^{even} content, first-order operators produce odd-zeta content—has a natural interpretation in the theory of mixed Tate motives over $\text{Spec}(\mathbb{Z})$. In this framework (Goncharov [34], Zagier [35]), the Riemann zeta value $\zeta_R(n)$ is a *period* of the mixed Tate motive $\mathfrak{M}(n) = \mathbb{Q}(-n)$ of weight n. Even zeta values $\zeta_R(2k) = \text{rational} \times \pi^{2k}$ are periods of motives that factor through the de Rham cohomology of even-dimensional compact manifolds (the Bernoulli formula). Odd zeta values $\zeta_R(2k{+}1)$ are periods of genuinely independent motives: they do not reduce to powers of π, and $\zeta_R(3)$ in particular is irrational (Apéry 1978) and conjectured algebraically independent from π.

The GeoVac operator-order discriminant maps onto this motivic partition:

Operator order	Motivic weight	Transcendental class	Example
2nd (Δ_{LB}, D^2)	even ($2k$)	$\zeta_R(2k) = \text{rat} \cdot \pi^{2k}$	κ, Vol(S^3), a_2
1st (D)	odd ($2k{+}1$)	$\zeta_R(2k{+}1) \perp \mathbb{Q}(\pi)$	D3: $\zeta_R(3)$ in

At motivic weight 3, $\zeta_R(3)$ appears as a period of $\mathbb{Q}(-3)$ in multiple independent geometric contexts: the Dirac Dirichlet series on S^3 (this work, Eq. 15), hyperbolic 3-manifold volumes (Goncharov [34]), two-loop QED $g{-}2$ (Petermann [37]; Sommerfield [38]), and Chern–Simons theory / algebraic K-theory ($K_5(\mathbb{Z})$ regulator). The common thread is S^3 or its quotients. Zagier's dimension conjecture [35] gives $d_3 = 1$: $\zeta_R(3)$ is the unique independent transcendental at weight 3, consistent with its single appearance in the GeoVac Dirac sector.

The connection to QED loop order is structurally predicted by the motivic correspondence. The T9 theorem (Eq. 21) guarantees that $\zeta_R(3)$ cannot enter through any single trace of a squared propagator D^{-2s}—i.e., at one loop. It enters at two loops, where irreducible products of first-order propagators $D^{-1} \otimes D^{-1}$ produce odd motivic weight. The operator-order discriminant thus determines the *loop order* at which new transcendental content first appears: one loop accesses only even-weight motives (π^{2k}); two loops access odd-weight motives ($\zeta_R(3)$, $\zeta_R(5)$, $\ldots$).

The even/odd Hurwitz spectral zeta of the first-order Dirac operator (Eq. 22 of Theorem 1) provides a sharp computational check. This exact formula separates the even and odd content at each s: at $s = 4$ both terms are π^{even} (Eq. 16: $D(4) = \pi^2 - \pi^4/12$); the first odd-zeta witness on the CH spinor bundle is Eq. (15): $D(5) = 14\,\zeta_R(3) - \frac{31}{2}\,\zeta_R(5)$. At even s the spectral sum is purely π^{even}; at odd $s \geq 5$, the $\zeta_R(s{-}2)$ and $\zeta_R(s)$ terms are independently transcendental. The QED vacuum polarization at one loop involves the Seeley–DeWitt coefficients a_0, a_1, a_2 of D^2, which by Eq. (21) are purely π^{even}; the Schwinger [36] result $a_e^{(1)} = \alpha/(2\pi)$ is a single power of π, consistent with one-loop = even motivic weight. At two loops, $\zeta_R(3)$ appears in the electron $g{-}2$ (Petermann [37]; Sommerfield [38]), consistent with the discriminant's prediction that odd-weight content requires irreducible products of first-order propagators.

This motivic interpretation refines the compactness thesis of Sec. I: exchange constants are not only projections from compact to non-compact geometry—they are *periods* of specific mixed Tate motives, and the operator order on the compact space determines the motivic weight of the resulting transcendental content.

h. Dirichlet L-values from vertex topology. The operator-order discriminant classifies free-propagator content. A third axis emerges when interaction vertices are included. The QED vertex on S^3 couples Dirac spinor harmonics at levels n_1, n_2 to a transverse vector harmonic at level n_γ with SO(4) selection rules requiring $n_1 + n_2 + n_\gamma$ odd (the γ^μ coupling flips parity). When this selection rule is imposed on the two-loop sunset diagram, it splits the Dirac Dirichlet series into even- and odd-n sub-sums with structurally different transcendental content. At $s = 4$ (the packing exponent), the exact Hurwitz decomposition gives:

$$D_{\text{even}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} - 4G + 4\beta(4), \tag{23}$$

$$D_{\text{odd}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} + 4G - 4\beta(4), \tag{24}$$

where $G = \sum_{n\geq 0}(-1)^n/(2n+1)^2 \approx 0.9159$ is Catalan's constant and $\beta(4) = \sum_{n\geq 0}(-1)^n/(2n+1)^4 \approx 0.9889$ is the Dirichlet beta function at 4. Both are values of the L-function $L(s, \chi_{-4})$ for the unique non-principal character mod 4. Crucially, G and $\beta(4)$ cancel in the full sum $D_{\text{even}} + D_{\text{odd}} = D(4) = \pi^2 - \pi^4/12$—the Dirichlet content is invisible without vertex restriction. The mechanism is the Hurwitz quarter-shift: even-n modes involve $\zeta(s, 3/4)$ and odd-n modes $\zeta(s, 5/4)$, whose difference is $4^s\beta(s)$ (a Dirichlet L-value), while their sum reduces to Riemann zeta (π^{even}).

We promote this to a theorem.

Theorem 2 (Three-axis classification of S^3 spectral transcendentals)**.** *The transcendental content of any spectral functional formed from the Camporesi–Higuchi Dirac operator D, the conformal scalar Laplacian Δ_{LB}, and a vertex coupling on round S^3 is determined by three independent axes:*

(A1) Operator order *(first vs. second). By Theorem 1, second-order operators (Δ_{LB}, D^2) produce π^{even} content exclusively; first-order operators (D) produce odd-zeta content $\zeta_R(2k+1)$ that is $\mathbb{Q}$-linearly independent of π.*

(A2) Bundle type *(scalar vs. spinor). Modulates rational coefficients, eigenvalue lattices, and degeneracies within the same transcendental class fixed by axis A1, but does not change the class. The (2nd-order, spinor) cell is degenerate with scalar calibration.*

(A3) Diagram topology *(free vs. vertex-restricted). Vertex parity selection rules from γ^μ couplings ($n_1 + n_2 + n_\gamma$ odd) project the Dirichlet sum onto quarter-integer Hurwitz shifts $\zeta(s, 3/4)$, $\zeta(s, 5/4)$ whose difference is $4^s\beta(s) = 4^sL(s, \chi_{-4})$. This introduces Dirichlet L-values—Catalan's constant $G = \beta(2)$, $\beta(4)$, and higher—that are absent from unrestricted free-propagator sums.*

Axes A1, A2, A3 are independent in the sense that no axis can be replaced or absorbed by the others: A1 is computational (operator → class), A2 is representation-theoretic (bundle → coefficient), A3 is interaction-theoretic (vertex → character). Output: the cell at coordinates (order, bundle, topology) *determines the transcendental class.*

Proof sketch. A1 is Theorem 1. A2 follows from the D^2 closed form (Eq. 21), where the spinor degeneracy $g_n^{\text{Dirac}} = 2(n+1)(n+2)$ produces only rational coefficient shifts relative to the scalar $g_n^{\text{scalar}} = (n+1)^2$ counterpart—both yield π^{even} at every integer $s \geq 2$. A3 follows from the Hurwitz quarter-shift identities $\zeta(s, 5/4) - \zeta(s, 3/4) = -4^s\beta(s)$ applied to the vertex-parity-restricted sum, giving Eqs. (23)–(24); the unrestricted sum is $D_{\text{even}}(s) + D_{\text{odd}}(s) = D(s)$, in which G and $\beta(4)$ cancel exactly, so the Dirichlet content is invisible without vertex restriction. Computational verification: PSLQ at 80-digit precision; 35/35 tests in `tests/test_qed_vertex.py`. □

i. Remark: Dirichlet L-tier as new transcendental class. The Dirichlet β-values introduced by axis A3 are a genuinely new transcendental class: they arise from the *interaction* (how electron and photon modes couple at the vertex), not from the free propagators. In the exchange-constant taxonomy of Sec. IV they constitute a *vertex exchange constant* tier living beyond both calibration π and Riemann odd-zeta. This is the entry that the RH-sprint χ_{-4} identity (Sec. II E, Eq. 7) places in closed form for the full Dirac Dirichlet difference.

j. Photon vector structure as a calibration exchange constant. The graph-native QED construction (Paper 28, GN-1 through GN-7) computes one-loop QED entirely on the finite Fock graph using exact algebraic arithmetic. On the graph, the photon lives on edges (1-cochains) as a *scalar* object—the edge Laplacian $L_1 = B^TB$ propagates it without reference to any vector structure. In the continuum, the physical photon is a transverse vector field on S^3 carrying $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ quantum numbers, and this vector structure enforces selection rules (vertex parity, SO(4) channel counting W) that are absent on the graph. Adding explicit photon angular momentum labels (q, m_q) with Clebsch–Gordan vertex coupling and E1 parity enforcement recovers 7 of 8 continuum QED selection rules for *scalar* electrons (Paper 28). The sole failure for scalar electrons is Furry's theorem: diagonal vertex couplings $V(a, a, q, m_q) \neq 0$ for $l \geq 1$ when q is odd. For *Dirac* electrons, Furry's theorem is recovered as well, achieving 8/8 (see below).

The calibration cost of the $1 \to 7$ transition is precisely one transcendental per loop: π, entering as the S^2 Weyl exchange constant $1/(4\pi)$ from the vector spherical harmonic normalization $\sqrt{(2l+1)(2q+1)(2l'+1)/(4\pi)}$. This is a concrete instance of the calibration exchange constants cataloged in the taxonomy table (Table I): the S^2 Weyl constant $\omega_2/(2\pi)^2 = 1/(4\pi)$ converts the discrete angular momentum quantum numbers into the continuous solid angle measure required for vector photon normalization. The entire scalar graph QED is π-free ($\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$); the vector photon introduces exactly the $\omega_2 = \pi$ entry from the Weyl constant table.

In the taxonomy, the photon's angular momentum structure therefore sits in the **calibration** tier, alongside the volume normalization $\mathrm{Vol}(S^3) = 2\pi^2$ and the Hurwitz-zeta regularization of infinite spectral sums: it is a property of the *manifold embedding*, not of the graph. This sharpens the graph/continuum partition for QED: the graph captures the algebraic/combinatorial content (rational propagators, algebraic CG vertices, π-free observables), while the continuum projection adds three calibration-tier ingredients: (1) volume normalization, (2) infinite-sum regularization, and (3) photon vector structure (cost: $1/(4\pi)$ per loop). The 8 continuum selection rules partition by transcendental cost into three tiers. For *scalar* electrons, the partition is 1–7–8: graph topology (1 rule, zero cost) $\to$ angular momentum conservation (6 rules, calibration cost $1/(4\pi)$) $\to$ field theory (1 rule, requires second-quantized C-symmetry). For *Dirac* electrons, the partition refines to 1–6–1–8: graph topology (1 rule, intrinsic, zero cost) $\to$ angular momentum conservation (6 rules, calibration $1/(4\pi)$) $\to$ Dirac spinor phase constraint (1 rule, intrinsic, zero cost).

k. The Dirac spinor phase constraint as an intrinsic tier. When the electron is described by a Dirac spinor $\psi = (f \cdot \Omega_\kappa,\ ig \cdot \Omega_{-\kappa})^T$ (Paper 23), the α-matrix vertex coupling is off-diagonal in the large/small component space. The factor of i on the small component makes the diagonal coupling $V(a, a, q, m_q)$ vanish identically for all states: $V_{LS} + V_{SL} = 0$ by Hermiticity. This is the mechanism by which Furry's theorem is recovered—tadpole diagrams (self-loops at vertex a) vanish because the single-particle Dirac vertex is kinematically off-diagonal.

The key taxonomic point is the *cost* of this recovery. The $1 \to 7$ transition (adding photon angular momentum labels) costs exactly π per loop—it is calibration content. The $7 \to 8$ transition (Dirac spinor phase) costs *nothing*: no new transcendental enters, no continuum embedding is required. The spinor phase constraint is a kinematic identity of the single-particle Dirac vertex, not a field-theoretic symmetry (C-conjugation). It lives in the same **intrinsic** tier as the graph topology rule (Gaunt/CG sparsity), not in the calibration tier.

This distinction has a concrete implication: the $7 \to 8$ transition does *not* require second-quantized field theory. Any single-particle framework using the Dirac equation on S^3 with vector-photon coupling automatically recovers all 8 selection rules. The standard attribution of Furry's theorem to charge conjugation symmetry is operationally correct in flat-space QFT but obscures the fact that, on the compact S^3 graph, the same selection rule has a simpler origin in the Dirac spinor structure itself.

l. Flat-space consistency of the transcendental classification. The three-axis taxonomy is derived from S^3 spectral sums at finite radius. A natural concern is whether the classification is a finite-curvature artifact. We verify that it is not. On S^3 of radius R, the Dirac eigenvalues scale as $|\lambda_n(R)| = (n + 3/2)/R$ while the degeneracies g_n are R-independent. The Dirichlet series therefore satisfies $D(s, R) = R^s\, D(s, 1)$ *exactly*, making the ratio $D(s, R)/R^s$ an R-independent structural invariant. In the $R \to \infty$ limit, Weyl's law confirms that the mode count $N(n_{\max}) = 2(n_{\max} + 1)(n_{\max} + 2)(n_{\max} + 3)/3$ matches the flat-space momentum-integral density $(2/3)\, R^3 p_{\max}^3$, verifying the spectral-to-momentum correspondence. The even/odd s-parity discriminant persists at every R: $D(s_{\mathrm{even}}, R)/R^s$ is π^{even}; $D(s_{\mathrm{odd}}, R)/R^s$ contains odd-zeta. The *same* transcendental constants—$\zeta_R(3)$, G, $\beta(4)$—that appear in the S^3 spectral sums also appear in flat-space two-loop QED (Petermann [37]; Sommerfield [38]), confirming that curvature modifies rational coefficients but not the transcendental class. An honest limitation: matching the full Petermann–Sommerfield coefficient $a_2 = -197/144 + \pi^2/12 + 3\zeta_R(3)/4 - \pi^2 \ln 2/2$ requires diagram-specific computation (renormalization subtractions, SO(4) Clebsch–Gordan algebra) beyond the scope of structural spectral sums. The flat-space limit thus validates the taxonomy as a *structural* property of the operator algebra, independent of the S^3 compactification scale.

m. Irreducibility of the Level 4 piecewise structure. The piecewise-smooth ρ-dependence of $\mu(\rho, R)$ at Level 4 is not a coordinate artifact: it cannot be resolved by geometric elevation to any higher-dimensional space (Track Z). Three approaches were ruled out: (i) algebraic blow-up of $|s - \rho|$ produces smooth sheets but with ρ-dependent integration limits ($\alpha = \arccos\rho$, $\arcsin\rho$), making matrix elements transcendental in ρ; (ii) Lie algebra elevation fails because $\min(s, \rho)/\max(s, \rho)$ is C^0 but not C^1, incompatible with matrix elements of any finite-dimensional Lie group representation (which are real-analytic); (iii) $S^3 \times S^3$ product space fails because two-center nuclear attraction cannot be smooth on any single compact manifold. The piecewise structure thus belongs to the embedding category in the taxonomy: it is the irreducible geometric content of placing two nuclei at finite separation in hyperspherical coordinates, analogous to the $1/r_{12}$ codimension obstruction (Track W) but affecting the nuclear rather than the electronic coupling.

n. Even-odd staircase and channel convergence rate. An extended convergence study at Level 4 (Paper 15, Track AA) reveals a pronounced even-odd staircase in D_e vs $l_{\max}$: even increments produce large gains (+42, +6.8, +1.7 pp) while odd increments produce near-zero gains (+0.66, +0.08 pp). The root cause (Track AC) connects directly to the exchange constant structure: the nuclear coupling matrix $\langle l_1' l_2' | V_{\rm nuc} | l_1 l_2 \rangle$ is diagonal in one electron's angular momentum index (Kronecker delta), so direct coupling from the ground channel $(0,0)$ requires a $\Delta l = 2$ transition enforced by the gerade constraint $l_1 + l_2 = $ even. Odd-$l_{\max}$ channels couple only indirectly through the electron-electron multipole expansion.

This selection rule constrains the *convergence rate* of the partial-wave expansion: the nuclear coupling—a piecewise exchange constant in the taxonomy—couples to the ground state only through quadrupolar ($\Delta l = 2$) transitions, producing an algebraic convergence rate $\sim (l + 1/2)^{-2}$ with an even-odd modulation. The σ and π channel sectors are completely decoupled (both nuclear and e–e coupling conserve m), with π contributing a constant $\sim$6.6 pp additive offset.

o. PK pseudopotential as composition exchange constant. The Phillips–Kleinman pseudopotential in the composed geometry (Paper 17) enforces core-valence orthogonality, which is a Pauli exclusion effect—distinct from the Coulomb exchange constants cataloged above. Without PK, no equilibrium geometry exists (the PES is monotonically attractive), establishing PK as essential for creating the bonding minimum. Ab initio derivation of the R-dependent PK scaling parameter $R_{\rm ref}$ from core properties fails due to a fundamental scale mismatch: all core-derived candidates (0.27–0.82 bohr) are too small by a factor of 4–11.

A complete survey of all solver $\times$ PK combinations (Paper 17, Table VI) confirms that PK is not an improvable approximation. The 2D variational solver eliminates $\sim$75% of the $l_{\max}$ drift (from the adiabatic approximation), but l-dependent PK has zero additional effect on the 2D solver. The residual drift (+0.15 bohr/$l_{\max}$) is intrinsic to the composed-geometry factorization.

Three graph-native alternatives to PK were investigated and all failed: (i) node exclusion between the core S^3 and valence Level 4 graphs has no coordinate correspondence; (ii) the fiber bundle Berry connection is intra-group, and the exchange operator P_{ij} mixes coordinate systems; (iii) spectral orthogonality (function-space) does not imply coordinate-space exclusion. The master obstruction is that antisymmetry requires a shared coordinate system; composition factorizes $\Psi_{\rm total}$ into incompatible coordinates.

PK therefore defines a sixth exchange type in the taxonomy: **composition constants** convert between a single-geometry (full N-electron) and a composed-geometry (factorized) description. Unlike the other exchange constants, which convert between discrete and continuous descriptions, composition constants measure the accuracy cost of the factorization itself. The "exchange rate" is quantified by the 144$\times$ angular compression that composed LiH achieves over a full 4-electron solver on S^{11} (Paper 17, Sec. V): the $\sim$5% $R_{\rm eq}$ error at $l_{\max} = 2$ is the irreducible cost of this compression.

This classification is validated by the full 4-electron Level 4N solver (Paper 17, Sec. VIII): exact S_4 antisymmetry produces molecular equilibrium without PK at $l_{\max} \geq 2$, confirming that PK approximates—rather than creates—the physics of Pauli exclusion. The composition exchange constant is the cost of replacing exact inter-group antisymmetry with a pseudopotential in the factorized coordinate system.

p. Inner-factor input data: a fourth tier disjoint from the outer-factor taxonomy. The exchange constants cataloged above all live on the GeoVac *outer* factor: the Fock-projected S^3 graph, its spinor lift, and their continuum limits. When the outer triple is combined with a Connes–Chamseddine *inner* finite spectral triple $T_F = (A_F, H_F, D_F, J_F, \gamma_F)$ to form an almost-commutative product $T = T_{\rm GV} \otimes T_F$ (Paper 32 § VIII.C; Sprint H1), the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ extends multiplicatively to the combined triple but with two structural reductions on the inner side.

Theorem 3 (η-trivialization on Krajewski-class finite triples)**.** *Let $(A_F, H_F, D_F, J_F, \gamma_F)$ be a finite even spectral triple in the Krajewski–Paschke–Sitarz lineage [18, 19]. The chirality grading axiom $\{\gamma_F, D_F\} = 0$ implies, for every odd $k \in \mathbb{N}$,*

$$\mathrm{Tr}\big(D_F^k \cdot e^{-tD_F^2}\big) \;\equiv\; 0, \qquad t > 0. \tag{25}$$

In particular, the M3 sub-mechanism (vertex-parity Hurwitz, $k = 1$) of the master Mellin engine vanishes identically on any Connes–Chamseddine-compatible inner factor.

Proof. $\{\gamma_F, D_F\} = 0$ and $\gamma_F^2 = 1$ make γ_F a unitary involution conjugating D_F to $-D_F$. Conjugation preserves traces, so $\mathrm{Tr}(D_F^k\, e^{-tD_F^2}) = \mathrm{Tr}((-D_F)^k e^{-tD_F^2}) = (-1)^k \mathrm{Tr}(D_F^k\, e^{-tD_F^2})$. Odd k forces zero. □ □

Theorem 4 (Multiplicative factorization of the combined spectral action)**.** *For the combined Dirac $D = D_{\rm GV} \otimes 1_F + \gamma_{\rm GV} \otimes D_F$ with $\gamma_{\rm GV}$ the outer chirality grading,*

$$D^2 \;=\; D_{\rm GV}^2 \otimes 1_F + 1_{\rm GV} \otimes D_F^2, \tag{26}$$

the cross term $\{D_{\rm GV} \otimes 1, \gamma_{\rm GV} \otimes D_F\} = (D_{\rm GV}\gamma_{\rm GV} + \gamma_{\rm GV} D_{\rm GV}) \otimes D_F$ vanishing by outer chirality anticommutation. Consequently

$$\mathrm{Tr}\big(e^{-tD^2}\big) \;=\; \mathrm{Tr}\big(e^{-tD_{\rm GV}^2}\big) \cdot \mathrm{Tr}\big(e^{-tD_F^2}\big), \tag{27}$$

and the master Mellin engine on the combined triple factorizes as (outer M_i ring) $\times$ (inner Yukawa Dirichlet ring).

The Mellin output of the inner factor is a generalized Dirichlet series in the eigenvalues $\{y_i\}$ of D_F (the Yukawa parameters, in the SM realization $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$):

$$\mathcal{M}\big[\mathrm{Tr}(D_F^k \cdot e^{-tD_F^2})\big](s) \;=\; \Gamma(s) \cdot \sum_i m_i \cdot y_i^{k-2s}, \qquad k \in \{0, 2\}, \tag{28}$$

with rational multiplicities m_i (including the SU(3) color factor 3 for the quark sectors). This output lies in the ring $\mathbb{Q}[y_1^{-2s}, y_2^{-2s}, \ldots, y_n^{-2s}]$ —a *parameter-tied Dirichlet ring* whose generators are free inputs to the framework, not transcendentals derived from any outer-factor mechanism.

We propose a sixth named tier in the taxonomy:

> **Inner-factor input data**: parameter-tied Dirichlet rings generated by the eigenvalues of a finite Dirac operator D_F on a Connes–Chamseddine-compatible inner factor. Source: the choice of finite spectral triple T_F and its Yukawa data. Type: empirical input from outside the framework, categorically disjoint from the intrinsic, conformal/calibration, embedding, algebraic-implicit, and composition tiers of the outer factor.

This tier sharpens the "no autonomous SM-distinguishing data" verdict of Sprints H1, G3, and G4a (Paper 32 § VIII.C): the Yukawa parameters, hypercharges, and mixing angles of the SM inhabit a ring with no overlap with any outer-factor ring, and GeoVac's taxonomy provides no derivation principle for them. The framework controls the outer factor's transcendental content via M1/M2/M3; the inner factor controls the SM-distinguishing parameters via Eq. (28); the two tensor cleanly by Theorem 4, and neither determines the other.

The structural separation is consistent with the universal/Coulomb partition of Paper 31 [16] read at the spectral-triple level: the universal sector (algebra A, selection rules) transfers freely; the potential-specific sector (Dirac operator $D_{\rm GV}$, calibration κ, α content) requires per-system rederivation; and now the inner sector (algebra A_F, finite Dirac D_F) requires empirical specification of its eigenvalue spectrum. Three structural sources of input data, three categorically disjoint tiers.

Computational evidence: the four-leg investigation (driver `debug/finite_spectral_triple_engine.py`, memo `debug/inner_factor_mellin_engine_memo.md`) verified Theorems 3–4 symbolically on the SM lepton sector, the full SM $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, and two comparators ($\mathbb{C}\oplus\mathbb{C}$ and $M_2(\mathbb{C})\oplus M_3(\mathbb{C})$). The KO-dim 6 cut on Krajewski candidates (required to complete the outer KO-dim 3 to the SM-compatible total $9 \equiv 1 \pmod 8$) and η-trivialization together leave a large candidate sub-class; the engine does *not* reverse-engineer A_F. This is a clean negative on the "second packing axiom" question: GeoVac's machinery says where A_F's Dirichlet ring sits in the taxonomy but does not pick out which finite spectral triple realizes it.

q. Open question: a second packing axiom for inner-factor data. The Paper 0 packing axiom forces the outer spectral triple's discrete labels (n, l, m, s) and the S^3 topology, but is silent on inner-factor content A_F. Sprint H1's $\mathcal{A}_{\rm GV} \otimes (\mathbb{C} \oplus \mathbb{H})$ extension produces a Higgs sector for any non-zero Yukawa Y, but no GeoVac data couples to that selection (the Sprint G3 commuting-$\mathbb{Z}_2$ theorem $\|\gamma_{\rm GV} \otimes I_F - I_{\rm GV} \otimes \gamma_F\|_{\rm op} = 2$ is the structural fingerprint). The Bertrand $\times$ Hopf-tower truncation provides only an upper bound on gauge content; the lower-bound forcing (why SU(2) and $U(1)$ appear alongside SU(3), why generation count = 3, why specific Yukawa ratios) is calibration-side. Whether a *sibling* axiom—distinct from the Paper 0 packing axiom—generates this data is an open question of the framework. Two candidate routes (charge-universal generation map; SU(5)/E_8/Spin(10) topological embedding) were tested in Phase 4H and falsified (Tracks SM-B, SM-C; CLAUDE.md §3). The honest scope statement is that GeoVac's machinery places inner-factor data in this taxonomic tier but does not derive it.

r. Thermal probes of the Yukawa Dirichlet ring. Sprint TD Track 3 (May 2026, `debug/sprint_td_track3_memo.md`) catalogues five thermal observables whose values depend on parameters of the inner-factor Yukawa Dirichlet ring of Eq. (28): lepton thermal free energy ($\Sigma\, y_i^2$ at $k = 2$); QCD thermal pressure $g_{\rm QCD}$ ($M_3(\mathbb{C})$ multiplicity ratio at $k = 0$); Higgs effective potential at finite T (off-diagonal CC block trace at $k = 2$); electroweak Debye masses ($(2 + N_{\rm gen})\, g^2$ at $k = 2$); and thermal running $\alpha(T)$ (rational charge sum and Yukawa-tied logarithmic cutoffs). Each observable's outer baseline factors cleanly through the Track 1 tensor-product spectral triple $T_{S^3} \otimes T_{S^1_\beta}$ (Sprint TD Track 1; Paper 35 tensor-product verification section), with the Stefan–Boltzmann $M_1 \times M_2 = -\pi^2/90\, T^4$ identity as the universal backbone. Each inner residual reduces to a generalized Dirichlet sum $\sum_i c_i\, y_i^{-2s}$ ($k = 0$) or $\sum_i c_i\, y_i^{2-2s}$ ($k = 2$) at integer s in the Yukawa eigenvalues. Three further candidates (axial chemical potential thermodynamics, parity-odd transport, finite-T EDMs) are filtered out by Theorem 3: they reduce to $k = 1$ traces on the inner factor and identically vanish, a structural prediction of the framework. None of the five surviving observables are predicted by GeoVac alone—they are consistency checks that constrain candidate T_F, not derivations. The headline empirical content is that the Marcolli–van Suijlekom gauge-network reading without Higgs ($Y = 0$) predicts no high-T electroweak symmetry restoration and is empirically falsified by the sphaleron transition at $T_C \approx 160$ GeV; this sharpens the positive-thin H1 verdict.

s. Chemistry-side analog: external atomic-physics calibration data. The same structural classification applies to GeoVac's composed natural-geometry chemistry side. The composed-qubit architecture builds heavy-atom molecules ($Z \geq 11$) using a frozen-core treatment: the [Ne], [Ar], [Kr], [Xe] cores are reduced to a screened $Z_{\rm eff}(r)$ profile via the FrozenCore solver, with hydrogenic plus multi-zeta valence orbitals built on top. The five Clementi–Raimondi 1963 exponents per row, the multi-zeta coefficients, and the screened-core orbital shapes are external atomic-physics inputs to the framework—they are not derived from any outer-factor Mellin mechanism (M1/M2/M3).

Sprint W1e period-class diagnostic (2026-06-04; `debug/sprint_w1e_period_class_memo.md`) tested whether the W1c $\to$ W1e chemistry-correlation wall sits in a higher-weight cyclotomic-mixed-Tate (M3) period class than the diagonal of the composed Hamiltonian. Eleven NaH correction terms extracted from sprints F4 / F5 / F6 were PSLQ-classified at 100 dps against M1, M2, M3, and an INNER analog basis built from the Clementi–Raimondi exponents. Result: zero outer-factor (M1/M2/M3) identifications across all eleven terms at audit ceiling 100 and permissive ceiling 10^6, after filtering for rational-only fits, basis-internal $\mathbb{Q}$-linear dependencies, and the curve-fit-audit failure mode (basis size vs. input precision). An independent random-rational null-hypothesis test against the same bases gave 0/50 false positives on M2 and M3, confirming the bases are sparse enough that "0 hits across 11 targets" is informative, not a measurement-noise artifact.

The PSLQ result is a consistency check, not a derivation: the load-bearing argument is structural. The M3 sub-mechanism lives at $k = 1$ in the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ (Sec. III E), which ties to vertex-parity / parity-character spectral sums on the GeoVac graph. The W1e correction terms have no vertex-parity content—they are FCI eigenvalues and Hartree integrals over hydrogenic plus multi-zeta orbital products—so they categorically cannot pick up M3 content. The PSLQ pass rules out the loophole "W1e might happen to land in M1/M2/M3 by coincidence"; the structural argument is what carries the negative.

The chemistry-side corrections classify cleanly into the chemistry-side analog of the inner-factor input-data tier: FCI eigenvalues are algebraic-implicit roots of polynomials with externally-fitted matrix elements; Hartree J and Slater–Condon K integrals are rational in external Z_{eff} values; Phillips–Kleinman barriers depend on the same external orbital fits. Six W1c / W1e closure attempts (CLAUDE.md §3 entries for cross-center PK, screened-Schrödinger valence, multi-zeta substitution, kernel-shape substitution, bonding-PK rank-1, basis enlargement, explicit-core Hartree) are explained at the structural level by this non-selection: they attempt outer-factor mechanisms (rational/algebraic modifications) on a wall whose content lies entirely in the external input-data tier.

The chemistry analog of Sprint H1's Yukawa non-selection theorem is the W1e wall: the framework can compose any Clementi–Raimondi exponent set into the heavy-atom Hamiltonian but cannot autonomously select the exponents. Outer-factor M-mechanism machinery is silent on inner-factor / calibration-data content in both the SM-side (Yukawa values) and the chemistry-side (atomic-physics calibration set) directions. The five chemistry-side observables in the Sprint TD Track 3 analog role (alkali-hydride binding energies, ionisation energies, fine-structure splittings, hyperfine splittings, polarisabilities) are not predicted by GeoVac alone; they are consistency checks that constrain candidate atomic-physics calibration sets, not framework-derived predictions.

t. Operational localization (Sprint R3-B falsifier, 2026-06-07). The W1e classification at the period-class level (0/11 hits in M1/M2/M3) is now sharpened by an operational test. Sprint R3-B exported the NaH balanced Hamiltonian to FCIDUMP at $n_{\max} \in \{2, 3\}$ via the `GeoVacHamiltonian.to_fcidump` interface and ran DMRG (sector-restricted FCI at FCI dim ≤ 784, which IS DMRG-at-infinite-χ) at every R in $[2.5, 6.0]$ bohr. The resulting PES is *bit-identical to the P4 direct-FCI baseline* (max diff $\sim 6 \times 10^{-13}$ Ha at every grid point), with no interior minimum and 21–24$\times$ overbinding at the experimental $R_e = 3.566$ bohr. Increasing $n_{\max}$ from 2 to 3 *deepens* the unphysical well rather than closing it. The wall is therefore not at the determinant-expansion level (DMRG cannot move it) and not at basis-truncation level ($n_{\max}$ sweep deepens it); it lives at the projection step from continuous Level 4 adiabatic+PK to second-quantized $(h_1, \mathrm{eri}, e_{\mathrm{core}})$ integrals. A parallel diagnostic on LiH composed and balanced (Sprint R3-A, same date) found the same monotone-descending qubit-FCI PES behaviour: the LiH 2.82% R_{eq} established at CHANGELOG v3.56.0 came from `ComposedDiatomicSolver.LiH_ab_initio` (continuous Level 4 multichannel adiabatic with PK), *not* from FCI on the qubit/FCIDUMP-exported integrals. The chemistry-side inner-factor input-data tier therefore has a specific structural home in the GeoVac construction: the projection step where continuous Level 4 multichannel information is reduced to the second-quantized integrals that seed qubit/composed solvers. This is the chemistry-side *operational* analog of the Sprint H1 Yukawa non-selection theorem at the projection level: the framework admits a chemistry Hamiltonian structurally but does not autonomously generate the calibration content of its $(h_1, \mathrm{eri}, e_{\mathrm{core}})$ tensors at this projection step.

V. CONNECTION TO THE FINE STRUCTURE CONSTANT

Paper 2 [7] observes that the fine structure constant α satisfies the cubic $\alpha^3 - K\alpha + 1 = 0$, where the coefficient

$$K = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right) = 137.036064 \qquad (29)$$

is constructed from spectral invariants of the Hopf fibration $S^1 \to S^3 \to S^2$. The combination rule $K = \pi(B + F - \Delta)$ is recorded as an Observation—an empirical formula with structural support but no first-principles derivation.

We observe that this combination rule has precisely the structure of a spectral-geometric exchange:

- $B = 42$ is discrete representation-theoretic data: the degeneracy-weighted SO(3) Casimir trace over the first three shells of S^3.

- $F = \zeta(2) = \pi^2/6$ is a continuous spectral invariant: the spectral zeta function of S^1 at $s = 1$.
- $\Delta = 1/40$ is a boundary correction mixing discrete eigenvalue data ($|\lambda_3| = 8$) with the cumulative state count ($N(2) = 5$).
- The overall factor π is $\omega_2 = \pi$, the 2D ball volume, which serves as the numerator of the S^2 Weyl density and as the packing-plane prefactor $\sigma_0 = \pi d_0^2/2$ in Paper 0 (the base of the Hopf fibration, the photon propagation sphere).

The structure of K is thus: the 2D ball volume $\omega_2 = \pi$ (the numerator of the S^2 Weyl density, and the same π that fixes Paper 0's packing-plane prefactor $\sigma_0 = \pi d_0^2/2$) multiplying a sum of the S^2 Casimir trace (B, discrete), the S^1 spectral zeta value (F, transcendental), and an S^3 boundary term (Δ, mixed). This is a Weyl-type formula that converts between the discrete bundle (representation labels on S^3) and the continuous coupling (the electromagnetic interaction strength measured in $\mathbb{R}^3$).

If this identification is correct, then the combination rule $K = \pi(B + F - \Delta)$ is not arbitrary numerology but an instance of the same spectral-geometric exchange that produces $\kappa = -1/16$ at Level 1 and $e^a E_1(a)$ at Level 2. The exchange constant for the Hopf bundle is K itself—the "cost" of projecting the S^3 electron state space through the S^1 gauge fiber onto the S^2 photon base, measured in units of coupling strength.

Observation 2 (Spectral-geometric reading)**.** *The combination rule $K = \pi(B + F - \Delta)$ has the shape of a spectral-geometric exchange constant for the Hopf fibration, analogous to the Weyl–Selberg constants for compact manifolds. Whether the specific form—π multiplying a sum of Casimir data and zeta values—admits a derivation from the spectral geometry of $S^1 \to S^3 \to S^2$ (e.g. via the heat-kernel expansion or the analytic torsion of the Hopf bundle) is an open question; no such derivation is known. The combination rule itself remains an Observation.*

Two additional observations support this identification:

Observation 3 (Structural match)**.** *The combination rule $K = \pi(B + F - \Delta)$ mixes discrete data ($B = 42$, from representation theory) with continuous spectral data ($F = \zeta(2) = \pi^2/6$, from the S^1 spectral zeta function), multiplied by an overall geometric factor ($\omega_2 = \pi$, the 2D ball volume that enters as the numerator of the S^2 Weyl density and as the packing-plane prefactor $\sigma_0 = \pi d_0^2/2$ in Paper 0, with S^2 the Hopf base). This is precisely the pattern identified throughout this paper: a transcendental conversion factor mediating between discrete and continuous descriptions of the same geometry. The structural match is evidence—not proof—that the combination rule is an instance of the exchange constant pattern rather than numerology.*

Observation 4 (α is likely transcendental)**.** *The fine structure constant α is a root of the cubic $\alpha^3 - K\alpha + 1 = 0$, where K involves π and $\zeta(2) = \pi^2/6$. Roots of polynomials with transcendental coefficients can be algebraic (e.g., $x^2 - \pi x + \pi(\pi - 1) = 0$ has the algebraic root $x = 1$), but this requires special cancellations that are generically absent. Since K is a non-trivial combination of π and π^2 with rational coefficients, it is overwhelmingly likely (though not proven) that α is transcendental—consistent with its role as a spectral-geometric exchange constant.*

A. Three-tier structure of the combination rule

An ongoing computational investigation into the origin of the three components of $K = \pi(B + F - \Delta)$—using exact-rational symbolic computation and high-precision arithmetic—has pinned down B and F as genuinely different objects on the Fock (n, l) lattice, and has eliminated a broad family of candidate mechanisms for Δ. We observe that the three terms appear to occupy three categorically distinct tiers of the exchange-constant taxonomy rather than specializing a single common generator. We summarize the four structural findings below; none of them alters the Observation-level status of Paper 2's cubic identity.

a. 1. A κ–B identity from the Fock weight. The Fock projection of the hydrogenic basis onto S^3 carries a residual momentum-space weight $w(p) = (p^2 + p_0^2)^{-2}$ after the overall $(2p_0)^{5/2}$ prefactor of the Fock map is stripped. Computation reveals that the diagonal matrix elements $\langle n, l|w|n, l\rangle_{p_0=1}$ for $n \le 3$ are exact rationals whose denominators divide $32 = 2\, d_{\max}^2$:

$$\{\langle n, l|w|n, l\rangle_{p_0=1}\}_{(n,l)} = \left\{\tfrac{7}{16}, \tfrac{5}{8}, \tfrac{3}{8}, \tfrac{5}{8}, \tfrac{17}{32}, \tfrac{11}{32}\right\} \tag{30}$$

for $(n, l) \in \{(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2)\}$. Weighted by the Hopf base factor $(2l + 1)\, l(l + 1)$ and summed,

$$\sum_{n=1}^{3}\sum_{l=0}^{n-1}(2l + 1)\, l(l + 1)\, \langle n, l|w|n, l\rangle_{p_0=1} = \frac{63}{4} = 6\, B\, |\kappa|, \tag{31}$$

with $B = 42$ the Casimir trace of Paper 2 and $\kappa = -1/16$ the calibration constant of Sec. VII F. Equivalently, averaging over the six (n, l) cells gives $B\, |\kappa| = 21/8$. This is the first exact rational algebraic link in the GeoVac framework between the calibration constant and the Hopf base invariant: $|\kappa| = 1/d_{\max}^2$ appears as the universal denominator factor of the Fock-weight matrix elements, while B appears in the Hopf-weighted sum. The identity (31) is exact-rational with no fitted parameters (the arithmetic closes: $9/4 + 51/16 + 165/16 = 63/4 = 6\, B\, |\kappa|$); the six Fock-weight values of Eq. (30) are stated here, and a regression test deriving them from the graph and checking the identity is a logged coverage gap (`docs/claim_test_matrix.md`).

b. 2. A Dirichlet identity for F. Define the Dirichlet series of the S^3 Fock shell degeneracy $g_n = n^2$,

$$D_{n^2}(s) \;:=\; \sum_{n=1}^{\infty} g_n\, n^{-s} \;=\; \sum_{n=1}^{\infty} n^{-(s-2)} \;=\; \zeta_R(s-2), \tag{32}$$

evaluated at the packing exponent $s = d_{\max} = 4$ of Paper 0:

$$D_{n^2}(d_{\max}) = \zeta_R(2) = \frac{\pi^2}{6} = F. \tag{33}$$

The exponent $s = 4$ admits three independent natural readings: $s = d_{\max}$ from Paper 0's packing axiom; $s = \dim(\mathbb{R}^4)$, the ambient Euclidean space into which S^3 is stereographically projected in Paper 7; and $s = 2N_{\rm init}$ from the same packing axiom. The weight n^2 is the degeneracy of the n-th S^3 shell in Fock's projection. To our knowledge (33) is the first identification of F from a graph-intrinsic quantity in the GeoVac framework—that is, F is produced here without invoking an embedding metric or a sphere-spectral construction. No other natural Dirichlet series on the shell lattice (per-shell Casimir, state count, Laplacian eigenvalue, l-max weight) produces an exact hit at any integer s in the range $[3, 12]$.

c. 3. Three independent spectral homes (Theorem 3). Combining the κ–B identity (31) with the Dirichlet identity (33) and the Paper 2 canonical form $1/\Delta = |\lambda_{n_{\max}}|\, N(n_{\max}-1)$, we promote the three-tier structure to a theorem; the additive combination rule itself remains an Observation.

Theorem 5 (Three independent spectral homes for B, F, Δ)**.** *Let $K = \pi(B + F - \Delta)$ be the combination of Paper 2 [7] with $B = 42$, $F = \pi^2/6$, $\Delta = 1/40$. The three components have structurally independent spectral origins on the Fock-projected S^3 graph:*

(B) $B = \sum_{n=1}^{3}\sum_{l=0}^{n-1}(2l{+}1)\, l(l{+}1) = 42$ *is a* finite *Casimir trace of the scalar* SO(3) *Laplacian, truncated at $n_{\max} = 3$; algebraic class: rational integer.*

(F) $F = D_{n^2}(s = d_{\max}) = \zeta_R(2) = \pi^2/6$ *is the* infinite *Dirichlet series of the Fock-shell degeneracy $g_n = n^2$ at the packing exponent $s = d_{\max} = 4$ of Paper 0; algebraic class: transcendental, in $\mathbb{Q}\cdot\pi^2$.*

(Δ) $\Delta^{-1} = |\lambda_{n_{\max}}|\, N(n_{\max}-1) = 8\cdot 5 = g^{\rm Dirac}_{n=3} = 40$ *is the* boundary *product at the truncation edge—equivalently the single-chirality Camporesi–Higuchi Dirac mode degeneracy at $n_{\rm CH} = 3$; algebraic class: rational, finite, on the* spinor *bundle.*

The three objects do not arise from a single common generator: B is a finite combinatorial sum, F is an infinite arithmetic series, Δ is a finite boundary degeneracy on a different bundle (spinor vs. scalar). Each lives in a distinct cell of the operator-order × bundle classification of Theorem 1 (B: 2nd-order / scalar / finite-truncation; F: 2nd-order / scalar / infinite-Dirichlet; Δ: 1st-order / spinor / boundary).

Proof sketch. For (B) and (F): Eqs. (31) and (33) above, by direct exact-rational computation. For (Δ): the Camporesi–Higuchi spectrum $|\lambda_n| = n + 3/2$ with degeneracy $g_n^{\rm Dirac} = 2(n+1)(n+2)$ gives $g_3^{\rm Dirac} = 40$ at the Paper 2 selection-principle cutoff $n_{\rm CH} = 3$. The scalar-Laplacian factorization $|\lambda_3| \cdot N(2) = 8 \cdot 5 = 40$ is a different presentation of the same integer; the spinor Dirac form $g_3^{\rm Dirac}$ is a single polynomial identity $2(n{+}1)(n{+}2)|_{n=3} = 40$, structurally cleaner. Independence: paragraph 4 below catalogues the negative results that exclude any common Dirichlet, sphere-spectral, or Hopf-fiber generator simultaneously producing B, F, and Δ. □

Observation 5 (Combination rule, restated)**.** *The additive combination $K = \pi(B + F - \Delta)$, with the three components characterized by Theorem 5, equals α^{-1} to 8.8×10^{-8}. No first-principles derivation of the combination is known. In the lattice gauge language of Paper 25 [10] and the non-abelian SU(2) extension of Paper 30 [14], K/π has the shape of a finite-cutoff spectral-action coefficient (matter trace + gauge-fiber zeta + Dirac boundary mode count), but the equality with α^{-1} is an empirical observation, not a theorem. The supertrace analysis on unit S^3 (Paper 28 [12], Sprint 2 Q-tracks) confirms that all three components are invisible to the perturbative Connes–Chamseddine asymptotic expansion—K/π lives in the non-perturbative sector if it lives in any spectral-action sector at all.*

d. 4. Negative results on candidate mechanisms. The three-tier reading is supported by the systematic elimination of alternative unifying mechanisms. Computation (exact rational symbolic arithmetic and `mpmath` at 50 decimal places) has ruled out, within the tolerance indicated: (i) the Hopf-twist spectral comparison S^3 versus $S^1 \times S^2$ (cleanest near-miss $\sim 6.8 \times 10^{-3}$); (ii) higher Casimir traces on S^3 at all orders through $n_{\max} = 3$; (iii) Hopf graph quotient Laplacian invariants for $S^3 \to S^2$ over the m-fiber; (iv) the discrete S^1 fiber spectral zeta (path-graph Laplacian, in four weighting schemes); (v) the continuous S^1 fiber via Mellin and heat-kernel expansions (the Jacobi inversion $\theta_{S^1}(t) \sim \sqrt{\pi/t}$ forces every order to be linear in π: the $\sqrt{\pi}\cdot\sqrt{\pi}\cdot\mathbb{Q} = \pi\cdot\mathbb{Q}$ structure collapses any expected $\zeta_R(2)$ contribution to π times a rational at every order); (vi) S^5 spectral geometry, including zetas, Seeley–DeWitt coefficients, and functional determinants (all rational, or reducible to $\zeta_R(\text{odd})$ and $\log 2$); and (vii) a ζ-combination scan for Δ over 3,228 candidates built from Riemann and Hurwitz zetas, Bernoulli numbers, and Stieltjes constants, with zero strict hits within 10^{-25} of $1/40$.

The structural reason for these negative results is that round-sphere Laplace–Beltrami operators produce (i) integer powers of π in their volumes, (ii) rational Seeley–DeWitt coefficients, and (iii) $\zeta_R(\text{odd}) + \log 2$ combinations in their spectral determinants—but never $\pi^2 = 2\zeta_R(2)$ additively next to a rational. This is consistent with the π-free certificate of the Bargmann–Segal lattice (Paper 24), which shows that the analogous lattice for the 3D harmonic oscillator on the Hardy space of S^5 is bit-exactly rational in exact arithmetic; and the same rigidity holds for S^5 Laplace–Beltrami spectral invariants generally.

e. Reframed open question. The derivation question "where does each piece of K come from?" has been partially answered: B from a finite Casimir truncation (with the κ–B link of Eq. 31), F from an infinite Dirichlet series at the packing exponent (Eq. 33), and Δ from a boundary product at the truncation edge (irreducible to ζ-content within the candidate set tested). The remaining structural question is no longer a derivation question about any single component: it is a question about why the *additive* combination $B + F - \Delta$, multiplied by the overall factor $\omega_2 = \pi$, equals α^{-1} to 8.8×10^{-8}. That is a question about the conspiracy of three categorically different objects—a finite sum, an infinite Dirichlet series, and a boundary product—rather than the specialization of a single generator. Paper 2's identification remains an Observation.

VI. OBSERVABLE CLASSIFICATION BY TRANSCENDENTAL CONTENT

The exchange constant taxonomy (Sec. IV) classifies the *constants themselves* by their mathematical origin. A complementary classification addresses the *observables*: given a physical quantity to be computed, what class of transcendental content does it require?

The conformal equivalence between the graph and continuous descriptions (Paper 7) implies a natural partition of observables into three classes, determined by what type of projection is needed to evaluate them.

1. **Class S (spectral observables):** Energy eigenvalues, degeneracies, selection rules, and transition amplitudes between eigenstates. These require only the graph eigenvalue spectrum and the calibration constant κ. No transcendental content beyond $\kappa = -1/16$ (rational) is needed.

2. **Class P (single-particle spatial observables):** Electron density $\rho(\mathbf{r})$, expectation values $\langle r^k \rangle$, momentum distributions, and any quantity requiring the wavefunction evaluated at a point in $\mathbb{R}^3$. These require the conformal projection $S^3 \to \mathbb{R}^3$, which introduces π through the volume element $d\Omega_{S^3} = 2\pi^2$ and the spherical harmonic normalization $\int |Y_{lm}|^2 d\Omega = 1$ (with $d\Omega$ carrying 4π).

3. **Class C (multi-particle correlated observables):** Correlation energies, adiabatic potential energy surfaces, multi-electron expectation values, and any quantity requiring the inter-particle coordinate. These require the higher exchange constants: $e^a E_1(a)$ at Level 2 for $m \neq 0$ states (Sec. II B), $\mu(R)$ at Level 3 (Sec. II C), $\mu(\rho, R)$ at Level 4 (Sec. II D), and the PK composition constant at Level 5.

This classification is stated as a structural claim:

Claim 4. *The class of transcendental content required to evaluate a quantum-mechanical observable is determined by the type of projection linking the graph description to the coordinate system in which the observable is defined. Spectral observables (Class S) require no transcendental content. Single-particle spatial observables (Class P) require the Weyl constants ($\pi^{d/2}$). Multi-particle correlated observables (Class C) require the embedding, flow, or composition exchange constants of Sec. IV.*

Claim 4 is falsifiable: for any observable, one can determine its class, identify the required exchange constants, and verify that no additional transcendental content enters the computation. The following worked examples illustrate each class.

A. Example 1: Hydrogen energy levels (Class S)

The hydrogen energy spectrum $E_n = -Z^2/(2n^2)$ is a spectral observable. The continuum S^3 Laplace–Beltrami operator that the graph Laplacian converges to has eigenvalues $\lambda_n = -(n^2 - 1)$, which are integers for all n (Paper 7, Eq. (12); Paper 0, Sec. II). The degeneracies $g_n = n^2$ are integers. The Fock conformal mapping (Paper 7, Sec. IV) converts these integer eigenvalues to the Rydberg spectrum via the energy-shell constraint $p_0^2 = -2E$ and the stereographic focal length:

$$E_n = -\frac{Z^2}{2n^2}\ \mathrm{Ha}, \tag{34}$$

with the calibration constant $\kappa = -1/16$ (rational) mediating between graph eigenvalues and physical energies (Sec. VII F). Every quantity in this derivation—eigenvalues, degeneracies, κ—is an integer or rational number. No transcendental content enters.

Selection rules provide a second example. The dipole selection rule $\Delta l = \pm 1$ follows from the triangle inequality on the Gaunt integral $\int Y_{l_1 m_1} Y_{1q} Y_{l_2 m_2}\, d\Omega$, which reduces to a product of Wigner $3j$ symbols—rational numbers computed from factorials of integers [1]. No transcendental content enters.

B. Example 2: Electron density (Class P)

The electron density $\rho(\mathbf{r}) = |\psi(\mathbf{r})|^2$ requires evaluating the wavefunction in position space. On the graph, the state $|n, l, m\rangle$ is a node with an integer label; in $\mathbb{R}^3$, the corresponding wavefunction is $\psi_{nlm}(\mathbf{r}) = R_{nl}(r)\, Y_{lm}(\theta, \phi)$. The conformal projection $S^3 \to \mathbb{R}^3$ (Paper 7, Sec. III) introduces the conformal factor $\Omega = 2p_0/(p^2 + p_0^2)$, which mediates between the unit-sphere measure $d\Omega_{S^3}$ and the flat-space volume element $d^3\mathbf{r}$.

The spherical harmonic normalization $\int |Y_{lm}|^2\, d\Omega = 1$ introduces π through the solid angle $\oint d\Omega = 4\pi$. The radial normalization $\int_0^\infty |R_{nl}|^2\, r^2\, dr = 1$ introduces no additional transcendentals for hydrogen (the integrals are rational multiples of a_0^{-3} times exponentials, evaluated in closed form).

The density is thus a Class P observable: it requires the π that enters through the projection from the dimensionless S^3 topology to the dimensionful $\mathbb{R}^3$ coordinate system.

C. Example 3: Helium ground state energy (Class C)

The helium ground state energy $E_0 = -2.9037$ Ha [25] is a correlated observable. The Level 3 hyperspherical solver (Paper 13) computes it via the adiabatic potential $U(R)$, which depends on the angular eigenvalue $\mu(R)$. The function $\mu(R)$ satisfies the characteristic polynomial $P(R, \mu) = 0$ (Sec. VII E), with coefficients in $\mathbb{Q}(\pi, \sqrt{2})$ inherited from the angular coupling matrix [23]. The energy is obtained by integrating the hyperradial Schrödinger equation with $U(R)$ as the effective potential—a definite integral over the algebraic curve, producing a transcendental number.

On the $S^3 \times S^3$ product space (Level 3 FCI), the same energy can in principle be computed without $\mu(R)$: the Hamiltonian matrix has entries given by Gaunt integrals (rational) and the Slater F^0 direct Coulomb integrals (algebraic on S^3, Paper 7 Sec. VI). The FCI result (0.35% error) demonstrates that the correlated energy *can* be approximated within the algebraic framework, but the $\mu(R)$ parameterization is needed to achieve sub-0.1% accuracy. The transition from Class S to Class C reflects the physical content: electron-electron correlation couples the two S^3 factors, and the coupling strength varies with the hyperradial coordinate R—a spatial degree of freedom that exists only in the projected description.

a. Three-layer decomposition (v2.6.0). The algebraic Casimir CI (Track DI Sprint 2) makes the exchange constant structure of the He ground state fully explicit. At each $n_{\max}$, the FCI matrix $H(k) = Bk + Ck^2$ has rational entries as a function of the orbital exponent k, so the variationally optimal energy E^* is an algebraic number—a root of a polynomial with rational coefficients. At $n_{\max} = 1$: $k^* = 27/16$, $E^* = -729/256$ Ha (exact rationals). This reveals a three-layer structure:

Layer	Content	Exchange type
Rational	Slater integrals, Gaunt coefficients	None (graph-intrinsic)
Algebraic	Optimal exponents at finite $n_{\max}$	Calibration
Transcendental	e-e cusp ($\alpha = \pi/4$ on S^5)	Embedding

The Fock energy-shell self-consistency $k^2 = -2E$, which is exact at Level 1 (one electron, one scale), overconstrains the two-electron problem: self-consistent k gives 5–13% error vs. 1.5–1.9% for variational k. This is the first instance where a single calibration constant is insufficient—the transition from calibration to embedding exchange constants corresponds to the physical content of electron-electron correlation.

D. Boundary cases

Two boundary cases clarify the classification:

- **Ionization energies** are spectral (Class S) when computed as eigenvalue differences $\Delta E = E_n - E_m$, which involve only κ and integer quantum numbers. However, photoionization *cross sections* require the dipole matrix element $\langle \psi_f | \mathbf{r} | \psi_i \rangle$, which is a spatial observable (Class P) because it evaluates the wavefunction at positions in $\mathbb{R}^3$.

- **Two-electron energies** computed via FCI on $S^3 \times S^3$ are formally Class S (graph eigenvalues of the product-space Hamiltonian), but achieve limited accuracy due to the product-basis completeness deficiency (Sec. II D). High-accuracy two-electron energies require the Class C exchange constants. The classification thus tracks both the type of observable and the accuracy target.

The classification does not assert that one class is more fundamental than another. The conformal equivalence (Paper 7) ensures that all three descriptions—graph, projected single-particle, correlated multi-particle—are mathematically equivalent. The classification identifies which exchange constants mediate between them.

VII. DISCUSSION

A. Transcendence and the graph design principle

Exchange constants are transcendental because they convert between incommensurable descriptions: a discrete count (eigenvalues, representation labels) and a continuous measure (volume, curvature, geodesic length). Weyl constants carry $\pi^{d/2}$, and π is transcendental by Lindemann's theorem. Not all functions in the hierarchy are transcendental: $\mu(R)$ satisfies a polynomial $P(R, \mu) = 0$ (Sec. VII E), making it algebraic in R, with transcendental content inherited from angular coupling coefficients [29]. The irreducible transcendentals are the *constants* that enter the coupling matrix, not the parameterized functions built from them.

When the computation stays on the graph, exchange constants are unnecessary. This suggests a design principle: *minimize coordinate projections* to minimize transcendental content. The ideal solver stays on the graph; every departure incurs a transcendental toll.

B. Exchange constants are computationally cheap

Conceptual complexity does not correlate with computational cost: κ is one multiplication; $e^a E_1(a)$ is one function evaluation seeding an $O(N^2)$ recurrence; $\mu(R)$ is a 10–30 dimensional matrix diagonalization per R-point. The hierarchy tracks curvature mismatch severity, not computational difficulty. At Level 2, identifying the Stieltjes seed and absorbing it into the basis *reduces* cost (eliminating quadrature) while improving convergence (Sec. II B).

C. The hierarchy of transcendence

Exchange constants increase in complexity with curvature mismatch: constant ($\kappa = -1/16$, Level 1) $\to$ single transcendental seed ($e^a E_1(a)$, Level 2) $\to$ algebraic function of one parameter ($\mu(R)$ satisfying $P(R, \mu) = 0$ over $\mathbb{Q}(\pi, \sqrt{2})$, Level 3) $\to$ piecewise-algebraic function of two parameters ($\mu(\rho, R)$, Level 4). This mirrors the mathematical hierarchy: Weyl constants are periods ($\pi = \int dx/(1 + x^2)$), Stieltjes seeds are exponential periods ($E_1(a) = \int_1^\infty e^{-at}/t\, dt$), and adiabatic eigenvalues are algebraic functions of the continuous parameter built from transcendental coefficients.

D. Structural parallels

The pattern parallels string compactification, where compact internal geometry determines low-energy couplings [33]. We note also that $\mathrm{SO}(4, 2) = \mathrm{Iso}(\mathrm{AdS}_5)$, the isometry group of five-dimensional anti-de Sitter space, is the conformal group of the boundary $\mathbb{R}^{3,1}$; the Fock projection $S^3 \to \mathbb{R}^3$ is a Euclidean section of this conformal boundary correspondence.

E. The algebraic curve refinement

The characteristic polynomial analysis [23] refines the exchange constant taxonomy by distinguishing between transcendental *constants* and algebraic *functions* of continuous parameters:

- **Truly irreducible transcendentals:** The embedding constants $e^a E_1(a)$ (Level 2, $m \neq 0$) and the angular coupling integrals c/π and $c\sqrt{2}/\pi$ (Level 3 and 4) are irreducible—they cannot be eliminated by any algebraic manipulation. These are the genuine "exchange constants" in the Weyl–Selberg sense.
- **Algebraic functions of transcendental data:** The adiabatic eigenvalues $\mu(R)$ and $\mu(\rho, R)$ are algebraic functions whose coefficients inherit transcendentals from the coupling matrix. The R-dependence (or (ρ, R)-dependence) is algebraic at every truncation: μ satisfies a polynomial equation $P(R, \mu) = 0$ of degree d equal to the matrix dimension. At $l_{\max} = 0$, P is linear and $\mu(R)$ is an explicit rational function of R (with transcendental coefficients). At $l_{\max} = L$, P has degree $L + 1$ in both R and μ.
- **Calibration constant:** $\kappa = -1/16$ is rational (Sec. VII F).

This refinement has a practical implication: the "transcendental toll" of the adiabatic parameterization is *inherited*, not *accrued*. Moving from the product space $S^3 \times S^3$ to hyperspherical coordinates does not introduce new transcendental content; it rearranges existing transcendentals (from the angular coupling) into an algebraic function of R. The toll is paid once, in the coupling matrix, and the R-dependence is algebraically determined.

This does not yet eliminate the computational cost of pointwise diagonalization: at $l_{\max} \geq 1$, the characteristic polynomial has degree ≥ 2 in μ and must be solved numerically at each R-point. But it does mean that the solver operates on a known polynomial, not an opaque transcendental function. Whether this algebraic structure can be exploited computationally—for instance, via resultant methods, homotopy continuation, or algebraic PES representations—is an open question.

This algebraic structure is specific to Level 3, where the angular Hamiltonian $H(R) = H_0 + R \cdot V_C$ is a linear matrix pencil. At Level 4, the split-region Legendre expansion of the nuclear charge function introduces piecewise-smooth ρ-dependence that breaks the pencil structure (Track S). The Level 4 angular eigenvalues are piecewise-algebraic but do not satisfy a global characteristic polynomial.

F. Two routes to the value of $\kappa = -1/16$

The original identification of $\kappa = -1/16$ (Paper 0) was empirical: the unique constant that maps GeoVac graph eigenvalues onto the Rydberg energy levels. Below we give two independent routes to its numerical value and classify their agreement. As recorded in Paper 0 and Paper 7, that the calibration scale coincides with a conformal-geometry quantity is an *Observation*—a numerical coincidence with no known derivation bridging the two—not a derivation of κ's calibration role from the geometry.

Graph Laplacian spectrum. The GeoVac lattice (Paper 0) discretizes S^3 with states $|n, l, m\rangle$ connected by angular transitions $m \leftrightarrow m \pm 1$ and radial transitions $n \leftrightarrow n \pm 1$. Interior nodes have degree $d_{\max} = 4$ (two angular, two radial neighbors). The graph Laplacian $\mathcal{L} = D - A$ is positive semidefinite; its maximum eigenvalue satisfies $\lambda_{\max} \leq 2d_{\max} = 8$, with the bound saturated in the continuum limit ($n_{\max} \to \infty$). Numerical verification confirms monotonic convergence: $\lambda_{\max}(n_{\max}{=}5) = 6.62$, $\lambda_{\max}(10) = 7.61$, $\lambda_{\max}(20) = 7.90 \to 8$.

Energy calibration. The graph Hamiltonian $H = \kappa\mathcal{L}$ (no node weights) has its ground state at energy $\kappa\lambda_{\max}$, since $\kappa < 0$ maps the largest graph eigenvalue to the lowest physical energy. Matching this to the hydrogen ground state:

$$\kappa\lambda_{\max} = E_H = -\tfrac{1}{2} \implies \kappa = \frac{-1/2}{8} = -\frac{1}{16}. \qquad (35)$$

Factor identification. The value $-1/16 = -(1/2)(1/8)$ decomposes into:

1. $1/2$: the non-relativistic kinetic energy normalization $E = p^2/(2m)$, or equivalently the energy-shell constraint $E = -p_0^2/2$;
2. $1/8 = 1/(2d_{\max})$: the inverse of the maximum graph Laplacian eigenvalue, determined by the vertex degree of the lattice discretization.

Heat kernel test. The Seeley–DeWitt coefficients on unit S^3 ($a_0 = 2\pi^2$, $a_1 = 2\pi^2$, $a_2 = \pi^2$) produce no ratio matching $1/16$; neither does the Weyl counting function $N(\Lambda) \sim \Lambda^{3/2}/3$ or the spectral zeta. The conformal coupling $c_3 = 1/8$ for $d = 3$ numerically equals $1/\lambda_{\max}$, but this is coincidental: c_3 is a Yamabe invariant while $1/\lambda_{\max}$ depends on graph vertex degree.

Fock weight derivation (Note added v2.26.1). The same value is derivable independently from the Fock projection itself, without reference to graph vertex degree. The $l = 0$ eigenfunctions on S^3 are Gegenbauer polynomials $C^1_{n-1}(\cos\chi)$ (equivalently, Chebyshev polynomials of the second kind $U_{n-1}(\cos\chi)$). Their three-term recurrence gives transition amplitudes of $1/2$ between adjacent shells, so the squared inter-shell coupling is

$$c^2(n,l) = \tfrac{1}{4}\left[1 - l(l{+}1)/(n(n{+}1))\right], \qquad (36)$$

with $c^2(n,0) = 1/4$ for all n (Paper 7, corrected 2026-09-03 from a printed $1/16$ prefactor). The quantity that coincides with $1/16$ is the inverse Fock Jacobian $1/\Omega^4(0)$, where $\Omega(0) = 2$ is the stereographic conformal factor at $p = 0$ (equivalently $c^2(n,0)/4$). The graph-degree derivation (Eq. 35) and the Fock weight route yield the same value: the graph's maximum vertex degree $d_{\max} = 4$ and the conformal factor $\Omega(0) = 2$ are both fixed by the unit-S^3 geometry the graph discretizes ($2d_{\max} = \Omega(0)^4 = 2^4/2 = 8$). That the calibration scale κ equals this geometric value is the Observation; no derivation links the two.

A notable special case: at $(n,l) = (4,3)$ the Casimir factor $2/5$ times $1/\Omega^4(0)$ gives $1/40 = \Delta$ (Paper 2), an Observation about a composite of two geometric quantities (the coupling itself is $c^2(4,3) = 1/10$), linking the Paper 2 cutoff to the Dirac degeneracy $g_3^{\text{Dirac}} = 40$.

Classification. In Table I κ sits in the "conformal" tier: its numerical value $1/16$ *coincides* with the conformal-geometry quantity $1/\Omega^4(0)$ of the Fock projection, a value any discretization of S^3 would share. The two routes (graph degree and conformal factor) return the same number because both are computed from the unit-S^3 geometry; that the calibration scale κ equals it is an *Observation* (a numerical coincidence, no known derivation bridge), consistent with the Observation-tier status of κ recorded in Paper 0 and Paper 7. The Z-universality noted in Paper 0 ($\kappa \to \kappa Z^2$ for any nuclear charge) reflects the Z-independence of the graph structure, with only the energy-shell scaling $E \propto Z^2$.

VIII. STRUCTURAL INCOMMENSURABILITY WITHIN GEOVAC OBSERVABLES

The exchange-constant taxonomy developed in the preceding sections has repeatedly encountered a distinctive pattern: single framework observables in GeoVac, when unpacked, resolve into sums or products of categorically distinct transcendental or spectral objects with *no common generator*. The pattern has been observed in three independently-derived three-axis classifications. We name it for future reference: *structural incommensurability within GeoVac observables.*

Observation 6 (Structural incommensurability)**.** *Across the framework, single-observable statements ("two-loop QED spectral sum on S^3", "$K = \pi(B + F - \Delta)$", "zero structure of the GeoVac zeta") have repeatedly resolved under structural analysis into sums or superpositions of objects from categorically different cells of the operator-order $\times$ bundle $\times$ topology classification of Theorem 2. The objects at each cell are what they are, not derivations of each other; the appropriate taxonomy is three-axis, not single-generator. The status is empirical at three documented instances; it may sharpen if further framework observables decompose the same way or if a bridge is found that fails cleanly.*

a. Instance 1 (QED on S^3, Paper 28). Perturbative QED sums on S^3 [12] produce three transcendental classes simultaneously: even-zeta content (π^{even}, from second-order Riemannian operators), odd-zeta content ($\zeta(3), \zeta(5)$, from first-order Dirac), and Dirichlet-L content ($G = \beta(2), \beta(4)$, from vertex topology). These three classes appear together in the two-loop sunset sum but are discriminated by Theorem 2's three axes (operator order $\times$ bundle type $\times$ vertex parity), and no single mechanism generates all three.

b. Instance 2 (α-combination decomposition). By Theorem 5, the observed combination rule $K = \pi(B + F - \Delta)$ has three components with structurally distinct spectral homes: $B = 42$ is a finite Casimir trace on the scalar Laplacian at $m = 3$; $F = \pi^2/6$ is the infinite Dirichlet series of Fock degeneracies at the packing exponent $d_{\max} = 4$; and $\Delta^{-1} = 40$ is the degeneracy $g_3^{\text{Dirac}} = 2 \cdot 4 \cdot 5$ of the third Dirac mode on the spinor bundle. No common generator produces all three; K assembles three structurally-independent spectral objects. In the gauge-structure language of Paper 25 [10] and the SU(2) Wilson extension of Paper 30 [14], K/π has the shape of a finite-cutoff spectral-action coefficient (matter trace + gauge zeta + Dirac boundary mode count).

c. Instance 3 (Weil dictionary on the GeoVac Hopf graphs, Paper 29). The Ihara-zeta side (combinatorial, closed non-backtracking walks) and the spectral-Dirichlet side (Dirac-eigenvalue Dirichlet series) of the same Hopf graph [13] give *opposite* verdicts. Zero spacing: Ihara Poisson (CV ≈ 1), spectral GUE (CV ≈ 0.35). Dirichlet-L content: Ihara absent (bipartiteness makes the χ_{-4} support disjoint from walk-length support); spectral present in the closed form of Eq. (7). Hilbert–Pólya compatibility: Ihara categorically excluded; spectral compatible at current sample precision. The two sides are not two views of the same object; they are two distinct framework observables computed from the same Dirac-S^3 data, with structural support that is disjoint in the bipartite-χ_{-4} sense.

d. Origin: Paper 24's Coulomb/HO asymmetry. The meta-pattern is consistent with—and arguably a consequence of—the Coulomb/HO asymmetry established in Paper 24 [9]: the graph is rational/integer on both bundles, but the projection structures (second-order Riemannian for the scalar, first-order complex-analytic for the HO holomorphic sector, Dirac for the spinor) are categorically different, and their transcendental projections onto physical coordinates do not interpolate. Paper 25 [10] sharpens this further: abelian U(1) Wilson–Hodge structure transfers verbatim from S^3 to the Bargmann–Segal S^5 graph, but non-abelian SU(3) on $(N, 0)$ SU(3) irreps fails (Clebsch–Gordan intertwiners are not group elements), making non-abelian gauge content Coulomb-specific. The universal/Coulomb partition is therefore three-layered, not two: (i) spectrum-computing role of L_0 (yes S^3, no S^5); (ii) calibration π (yes S^3, no S^5); (iii) Wilson physical content (full S^3, reduced S^5).

e. Operational consequence. For taxonomy construction (Sec. IV), the consequence is that multi-component observables should be decomposed and each component classified separately, rather than the whole observable forced into a single tier. This changes nothing about Sec. VI's S/P/C classification of atomic observables (which are fine as they stand at the observable level), but it does mean that composite structural results—such as Paper 2's K or Paper 28's two-loop sunset—should be broken into their components before taxonomic assignment.

IX. CONCLUSION

The exchange-constant taxonomy of Sec. IV now admits three theorems and one observation, alongside the catalogued constants of Levels 1–5:

a. Theorem 1 (operator-order discriminant). On round S^3, the squared Dirac operator D^2 has spectral zeta $\zeta_{D^2}(s) = 2^{2s-1}[\lambda(2s-2) - \lambda(2s)]$ which is a polynomial in π^2 at every integer $s \geq 2$ (Eq. 21); the first-order Dirac operator has spectral zeta containing odd-zeta content $\zeta_R(2k+1) \perp \mathbb{Q}(\pi)$ at odd s (Eq. 22). Operator order is the primary transcendental discriminant; bundle type modulates rational coefficients only. Loop-order prediction (verified): no odd-zeta at one loop (Schwinger [36], $a_e^{(1)} = \alpha/2\pi$); $\zeta_R(3)$ at two loops (Petermann [37]; Sommerfield [38]).

b. Theorem 2 (three-axis classification). The transcendental content of any spectral functional on round S^3 is determined by three independent axes: (A1) operator order (1st vs. 2nd) via Theorem 1; (A2) bundle type (scalar vs. spinor) modulating coefficients; (A3) diagram topology (free vs. vertex-restricted) introducing Dirichlet L-values ($G = \beta(2)$, $\beta(4)$) through the quarter-integer Hurwitz shift. The classification is R-independent: $D(s, R)/R^s$ is a structural invariant, and Weyl's law preserves the even/odd-s discriminant in the flat-space limit.

c. Theorem 3 (three independent spectral homes for K). The three components of Paper 2's $K = \pi(B+F-\Delta)$ have structurally independent spectral origins: B a finite Casimir trace on the scalar Laplacian at $n_{\max} = 3$; $F = D_{n^2}(d_{\max}) = \zeta_R(2)$ an infinite Dirichlet series of Fock degeneracies at the packing exponent; $\Delta^{-1} = g_3^{\text{Dirac}} = 40$ a finite Dirac mode degeneracy on the spinor bundle. No common generator produces all three. The combination rule $K = \alpha^{-1}$ (Observation 5) remains a numerical observation without first-principles derivation.

d. Observation (structural incommensurability). The three theorems share a meta-pattern: framework observables that look unitary at the surface ("$K = \pi(B + F - \Delta)$"; "two-loop QED on S^3"; "zero structure of the GeoVac zeta") resolve under structural analysis into objects from different cells of the operator-order × bundle × topology grid. Observation 6 catalogues three independent instances; no further unification is currently visible.

e. Verified and open. The taxonomy has been validated across the full GeoVac hierarchy through Level 5; Claim 4 (transcendental content determined by projection type) has held across all tested cases. What remains open: (i) whether the three-axis classification of Theorem 2 predicts the physical two-loop coefficient (not just the transcendental class); (ii) the combination rule $K = \pi(B + F - \Delta)$ of Observation 5, whose component decomposition is now Theorem 5 but whose additive conspiracy remains unexplained; (iii) whether the structural incommensurability of Observation 6 sharpens to a theorem when further composite observables are decomposed.

f. Design principle. Staying on the graph eliminates exchange constants by eliminating projections. The program's effort to algebraicize solvers (Paper 12's Neumann expansion, Laguerre recurrences, Gaunt replacements, Paper 19's cross-center V_{ne} via incomplete gamma functions) is, in retrospect, an effort to eliminate exchange constants. The algebraic curve result (Sec. VII E) shows this has been more successful than previously recognized.

g. Note added v2.6.0. The He exchange constant census (Track DI) validates the taxonomy computationally. Graph-native FCI achieves 0.19% at $n_{\max} = 7$ with zero free parameters. Three-layer structure: (1) rational (graph h_1 + Slater V_{ee}, 98.6%); (2) topological (graph off-diagonal $T_\pm$, calibration κ, dominant); (3) transcendental (e-e cusp, embedding, ∼0.14% floor). FCI basis invariance confirmed: the floor is embedding content (cusp), not basis mismatch.

h. Note added v2.26.1 (Seeley–DeWitt cancellation). On unit S^3, the Seeley–DeWitt coefficient ratio $a_k^{D^2}/a_k^{\Delta_{\rm LB}} = 4 = \dim(S)$ at every order k, where S denotes the spinor bundle. This ratio is exact: the asymptotic mode density of the Camporesi–Higuchi Dirac spectrum is $g_n^{\rm Dirac} \sim 4n^2$, four times the scalar density n^2, and the SD expansion depends only on this asymptotic density. As a consequence, the perturbative Connes–Chamseddine supertrace $\mathrm{Str}[f(D^2/\Lambda^2)] \equiv \mathrm{Tr}_S[f] - \frac{1}{4}\mathrm{Tr}_D[f]$ vanishes identically in the asymptotic expansion. The three ingredients of $K/\pi = B+F-\Delta$ are all invisible to the SD expansion: B is a finite Casimir trace, F is an arithmetic Dirichlet series, and $\Delta^{-1} = 40$ is the Euler–Maclaurin upper boundary term of the Dirac mode-count sum at $n_{\rm CH} = 3$. The sign on Δ in the combination rule $K = \pi(B + F - \Delta)$ is the standard $(-1)^F$ grading: B and F are scalar-sector (bosonic) quantities that enter with $(+)$, while Δ is a spinor-sector (fermionic) quantity that enters with $(-)$. K/π lives entirely in the non-perturbative sector of the spectral-action supertrace on S^3.

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